

Rationalizing Tariff Structures in the Indian Downstream Aluminium Industry

Addressing the Inverted Duty Structure to Maximize Value Addition and Employment

Abstract

This report argues for rationalising the basic customs duty on primary aluminium (HS 7601), on the ground that the present structure taxes conversion rather than protecting it. The case rests on two independent bodies of evidence.

*The first is an estimate of effective protection. Applying a modified Corden measure to Supply–Use data and partner tariff schedules, we find that the thirteen downstream product lines HS 7604–7616 carry **negative** effective protection against Malaysia, Korea, Japan and Thailand by 2024–25, against a non-preferential benchmark of +0.50. Preferential agreements have removed the duty from the finished good while leaving it on the metal, so the escalation that ordinarily protects processing instead penalises it. Because bought-in inputs are 83 per cent of downstream output value, the resulting wedge is amplified roughly fivefold by the time it reaches value added.*

*The second is the Annual Survey of Industries. Partitioning the ASI on the same boundary the tariff schedule uses, we find that a rupee of value added created in downstream fabrication sustains **3.1 times** the employment of a rupee created in primary metal—a result that holds in every one of the twelve clean years, with a median of 3.9. Downstream converts **38.4 per cent** of value added into wages against 17.6 per cent upstream, and supports **151 persons per crore** of estimated corporate tax against 39. Within that aggregate, the single class most exposed to duty-free finished imports—structural products, which make HS 7610—grew nominal value added by 15 per cent in eleven years while its workforce did not grow at all. The association is consistent in sign throughout; we do not claim it is an identified causal effect, and say so.*

These findings are set against a demand environment that makes the distortion more costly each year. Public capital expenditure of INR 12.2 lakh crore in 2026–27, of which INR 2.93 lakh crore is railway capital outlay, generates demand for extrusions, structural sections and stranded conductor—precisely the lines on which domestic effective protection is negative. Compounding this, geopolitical chokepoints, subsidised Chinese competition and the carbon-compliance regimes now taking effect fall hardest on the segment least able to absorb them. The policy imperative follows: correct the input duty, and the demand that public investment is already creating can be met from domestic capacity rather than imported.

Contents

1 Introduction	3
1.1 Aluminium as a Strategic Metal	3

1.2 Strategic Importance for the Indian Economy	3
1.3 Scope and Objective: Focus on HS Codes 7601–7616	5
1.4 The Aluminium Value Chain	6
2 International Trade Dynamics	7
2.1 Supply-Chain Fragility: Geopolitical Chokepoints, War-Risk Insurance, and Energy Costs	10
3 Effective Rate of Protection and the role of Inverted Duty Structure	12
3.1 The Current Tariff Regime, Import Volumes, and the Role of FTAs	12
3.2 The Case for Countervailing and Anti-Dumping Measures against Subsidised Imports	13
3.3 Empirical Evidence: The Erosion of Effective Protection, 2015–16 to 2024–25	14
3.4 The “Lose-Lose” Scenario for the Indian Consumer	17
4 Policy Plight of MSMEs in the Aluminium Industry	18
4.1 Measuring the Sector: Where the Value Chain Divides	18
4.2 Value Added and Employment	19
4.3 Labour Intensity: The Central Finding	20
4.4 Labour Share: Where the Value Added Goes	21
4.5 Where the Pressure Falls: Structural Products	22
4.6 Employment Composition	24
4.7 Corporate Tax: Employment Yield and Revenue Stability	24
4.8 The Two Segments at a Glance	25
4.9 Compliance Headwinds: CBAM, ESG Norms, and the CAPEX Barrier to Exports	26
5 Conclusion	27
A Appendix: Technical Framework of ERP Analysis	31
A.1 Modified Corden’s Approach	31
B Appendix: ASI Measurement — Segmentation Rule, Formulas and Assumptions	32
B.1 The segmentation rule	32
B.2 Notation	32
B.3 Estimators	33
B.4 Construction of each figure	33
B.5 Two derived results used in Section 4	34
B.6 Apportionment, and why it does not affect the results	35
B.7 Assumptions	36
B.8 Robustness	37
C Appendix: Supplementary Figures	39

1 Introduction

1.1 Aluminium as a Strategic Metal

Aluminium is the third most abundant element in the Earth's crust—accounting for roughly 8%—and the single most abundant metal, though its reactive nature means it occurs in nature only in bound alumino-silicate forms rather than as free metal [4]¹. Over the past six decades global aluminium consumption has grown more than twenty-fold, one of the fastest expansions of any industrial metal. A distinctive combination of properties—light weight, high electrical and thermal conductivity, corrosion resistance, formability, and near-infinite recyclability—has made it the second-most consumed metal in the world after steel, with global consumption of approximately 97.8 million tonnes in 2023 [4]². Having overtaken copper in 1954 to become the most-produced non-ferrous metal, aluminium today accounts for more output than all other non-ferrous metals combined—some 108 MT in 2022, against roughly 26 MT for copper.

Often described as the “metal of the future,” aluminium underpins a remarkably broad set of end-uses: automobiles and surface transport, building and construction, packaging, electrical transmission, machinery and equipment, aerospace and defence, and consumer durables [4]³. Its low weight and recyclability place it at the centre of the global low-carbon transition—as a lightweighting material for electric vehicles, a structural input for renewable-energy infrastructure, and an enabler of energy-efficient buildings—while its infinite recyclability aligns naturally with circular-economy principles. These attributes have led governments worldwide to treat aluminium as a strategic industry: China formally designates it a “pillar industry,” the United States classifies it among the minerals critical to national defence and energy security, and the European Union, Russia, Canada, South Korea and the Gulf economies have each built dedicated industrial strategies around the metal.

1.2 Strategic Importance for the Indian Economy

For India, the strategic significance of aluminium spans economic growth, infrastructure development, defence, transportation, sustainability, and global competitiveness. As of the financial year 2024, India's installed primary aluminium capacity stood at 4.17 million tonnes, making it the second-largest primary producer in the world. Yet India's per capita consumption remains remarkably low—roughly 3.5 kg (FY24) against a global average of about 12 kg, and only a fraction of the levels seen in mature markets such as South Korea (42.8 kg, the world's highest), Bahrain (32 kg) and Germany (31 kg) [4]⁴. This gap signals an enormous, largely untapped domestic growth runway. Propelled by rapid urbanization, electrification of the grid, and the transition to electric mobility, domestic demand is projected to surge to 8.5 MT by FY30 and 28 MT by FY47 [4]⁵.

These aggregate projections translate into an exceptional, and specifically downstream, opportunity. The bulk of India's incremental demand to FY47 is expected to arise not from primary metal but from higher-intensity and new-to-India value-added applications—an estimated 15–18 MT from deeper penetration of existing uses (defence and civil-aviation airframes, packaged food and beverage, and the like) and a further 4–6 MT from newer applications such as greater aluminium content in automobiles and

¹Ministry of Mines, Government of India. (2025). *Vision Document on Aluminium Metal for India*.

²Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

³Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

⁴Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

⁵Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

transport—precisely the product categories fabricated by downstream MSMEs [4]⁶. On the trade side, India today remains a fringe player in the roughly 67 MT global aluminium market, holding a share of just 3.8% against China's 12.6%; national strategy targets lifting this to about 10% by FY47 while deliberately reorienting the export mix towards value-added downstream products rather than raw metal, and maximising the conversion of primary output into downstream goods by 2035 [4]⁷. Set against India's broader ambition of becoming a US\$30 trillion economy, capturing even a fraction of this trajectory would require the downstream sector to expand several-fold—underscoring why the structural disadvantages it currently faces carry consequences far beyond the firms directly affected.

The demand is already being created, by the public exchequer. These projections are not distant forecasts contingent on private appetite. A substantial part of the incremental demand is being generated now, by budgeted public expenditure, and it lands with unusual precision on the product lines this report is about. Capital expenditure in the Union Budget for 2026–27 is set at INR 12.2 lakh crore, of which **INR 2.93 lakh crore** is capital outlay on the railways—a record, and a 5.4 per cent increase on the preceding year—including INR 52,109 crore on rolling stock alongside seven new high-speed corridors and a further dedicated freight corridor [24]⁸.

Each of those heads converts into aluminium, and into specific tariff lines. Rolling stock is the clearest case: aluminium car bodies are at least 30 per cent lighter than stainless-steel equivalents, which is why the next generation of Indian trainsets is moving to aluminium construction, and why an order for one hundred aluminium-bodied Vande Bharat sleeper trainsets has been placed. A metro coach absorbs of the order of six to eight tonnes of *extruded profiles*—HS 7604—so that a single metro line of thirty six-coach trains represents something near 1,500 tonnes of extrusion on its own [27]⁹. The grid tells the same story through a different line: aluminium is the preferred conductor for overhead transmission and distribution, and the National Electricity Plan requires inter-regional transfer capacity of 143,850 MW by 2026–27 against a build-out of some 470 GW of solar and wind over the decade [25]¹⁰—demand which materialises as stranded wire and cable, HS 7614. Construction and infrastructure, meanwhile, remain the largest single end-use of Indian extrusions at roughly 280,000–300,000 tonnes, arriving as structural sections, doors, windows and façades: HS 7610.

The significance of this is not that demand is growing. It is *where* the demand falls. Extrusions (HS 7604), structural products (HS 7610) and stranded conductor (HS 7614) are among the lines on which Section 3 finds effective protection to be negative, and HS 7610 is the line that the industrial statistics of Section 4.1 show to be stagnant. India's aluminium extrusion consumption is projected at **858,000 tonnes in 2026**, up 7.9 per cent in a single year [26]¹¹. The question a tariff schedule has to answer is who supplies that growth. On the present structure, a fabricator buying metal at a duty-inclusive price competes for this business against a finished import entering at zero duty—so public investment intended to build domestic industrial capability is, at the margin, underwriting fabrication capacity abroad. That is a strange result for an

⁶Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

⁷Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

⁸Ministry of Finance, Government of India. *Union Budget 2026–27*, Expenditure Profile and Demands for Grants; railway allocations as presented on 1 February 2026.

⁹Trade estimates; Titagarh's Pune Metro coaches are India's only operational aluminium metro fleet. Figures are indicative of order of magnitude rather than precise engineering specifications.

¹⁰Central Electricity Authority, *National Electricity Plan, Volume II: Transmission* (2024), Ministry of Power, Government of India.

¹¹"India's aluminium extrusion demand nears 858,000 tonnes," *AL Circle* (2026); 795,000 tonnes in 2025.

expenditure programme of this size to produce, and it is correctable at the tariff line rather than through further subsidy.

Whether any of this potential is realised depends, above all, on the downstream sector. Comprising over 3,500 Micro, Small, and Medium Enterprises (MSMEs), it is the primary engine of value addition, employment, and product innovation in the aluminium economy—and, as this report argues, the segment most exposed to the structural tariff anomalies examined in the sections that follow.

Two things about this downstream segment explain why it is so sensitive to policy. The first is that, unlike the highly consolidated and capital-intensive upstream smelting industry, downstream fabrication is geographically dispersed, fragmented, and dominated by small firms operating on structurally thin margins, leaving it far more exposed to external cost shocks [15]¹². The second is that the price of its main raw material is not set at home at all. It is tethered to a global benchmark that is itself highly volatile: empirical analysis of non-ferrous metals finds strong—and, since the financialisation of commodity markets, intensifying—spillovers from geopolitical risk and global economic-policy uncertainty into aluminium prices [14]¹³. So the segment that adds the most value and creates the most jobs is also the one least sheltered from forces well outside its control. Tariff design has, so far, largely ignored that asymmetry.

Nothing throws this asymmetry into sharper relief than the contrasting financial health of the two segments. India's consolidated primary producers are not fragile entrants in need of tariff protection but highly profitable incumbents. Brokerage analysis of the country's fastest-expanding primary producer, for instance, projects operating earnings (EBITDA) compounding at roughly 29% a year over FY26–FY28 to around INR 419 billion (about US\$4.4 billion), even as its hot-metal cost of production falls from approximately US\$1,914 per tonne to below US\$1,600 and its net debt is all but extinguished [12]¹⁴. Set against operating surpluses of that size, the 7.5% basic customs duty on primary metal protects producers who are already thriving—which is to say, it protects very little. What it does in practice is transfer margin from a thin-margin downstream sector to a comfortably profitable upstream one. Raising input duties further would therefore shield nothing that genuinely requires shielding, whereas rationalising them downward would pass through directly to the competitiveness—and, for many firms, the survival—of the downstream fabricators on which the sector's value addition and employment depend.

1.3 Scope and Objective: Focus on HS Codes 7601–7616

The objective of this report is to advocate for the rationalization of tariff lines for input products utilized in the downstream aluminium categories (HS Codes 7601–7616). We argue that the current tariff structure disproportionately harms domestic players and must be realigned to support domestic manufacturing and value chain integration. This is not a novel position; it rests on a body of existing scholarship. India-focused research has argued that lowering import duties on primary aluminium—particularly low-carbon metal from producers such as Canada, Norway and Iceland—would normalise domestic input prices and directly raise the competitiveness of the downstream sector, including

¹²*Comprehensive Economic Analysis of the Global Downstream Aluminium Industry*, 2026 (synthesis of trade data and peer-reviewed literature). [15].

¹³Ben-Salha, O., Zmami, M., Waked, S. S., Najjar, F., & Alenazi, Y. M. (2025). *On the time-varying spillover between nonferrous metals prices, geopolitical risks, and global economic policy uncertainty*. *Economic Change and Restructuring*, 58(5). [14].

¹⁴“Vedanta Aluminium to rally 22%? Nuvama initiates ‘Buy’ with 29% EBITDA forecast,” *AL Circle*, 14 July 2026.

its capacity to export value-added products to carbon-regulated markets [13]¹⁵. More broadly, the trade-and-development literature establishes that access to competitively priced and varied intermediate inputs is a principal driver of domestic product growth and innovation—evidence drawn, notably, from India’s own experience with input-tariff liberalisation [2]¹⁶.

1.4 The Aluminium Value Chain

As illustrated in Figure 1.1 (reproduced in Appendix B), the aluminium manufacturing process involves three distinct phases, spanning from raw extraction to advanced fabrication.

- **Phase 1 (Upstream):** The cycle begins with the mining of bauxite ore, which is refined into alumina and subsequently smelted into unwrought primary aluminium (HS 7601). Recycled scrap (HS 7602) also enters the metallurgical cycle at this stage.
- **Phase 2 (Midstream / Downstream Genesis):** This unwrought metal is cast, extruded, rolled, or drawn into essential semi-finished goods such as bars, plates, sheets, foils, and wire (HS 7604–7609).
- **Phase 3 (Downstream):** These intermediate inputs are transformed into specialized, highly value-added end products, including architectural structures, packaging containers, household utensils, and electrical cables (HS 7610–7616).

This report specifically addresses the structural trade imbalances occurring as materials move from Phase 1 into Phases 2 and 3.

¹⁵Arora, A. (2023). *Assessing the Impact of CBAM and Tariffs in the Indian Aluminum Industry*. IIFT Working Paper Series, WPS No. EC-23-65, Indian Institute of Foreign Trade, New Delhi. [13].

¹⁶Goldberg, Khandelwal, Pavcnik & Topalova (2010), *Imported Intermediate Inputs and Domestic Product Growth*. [2].

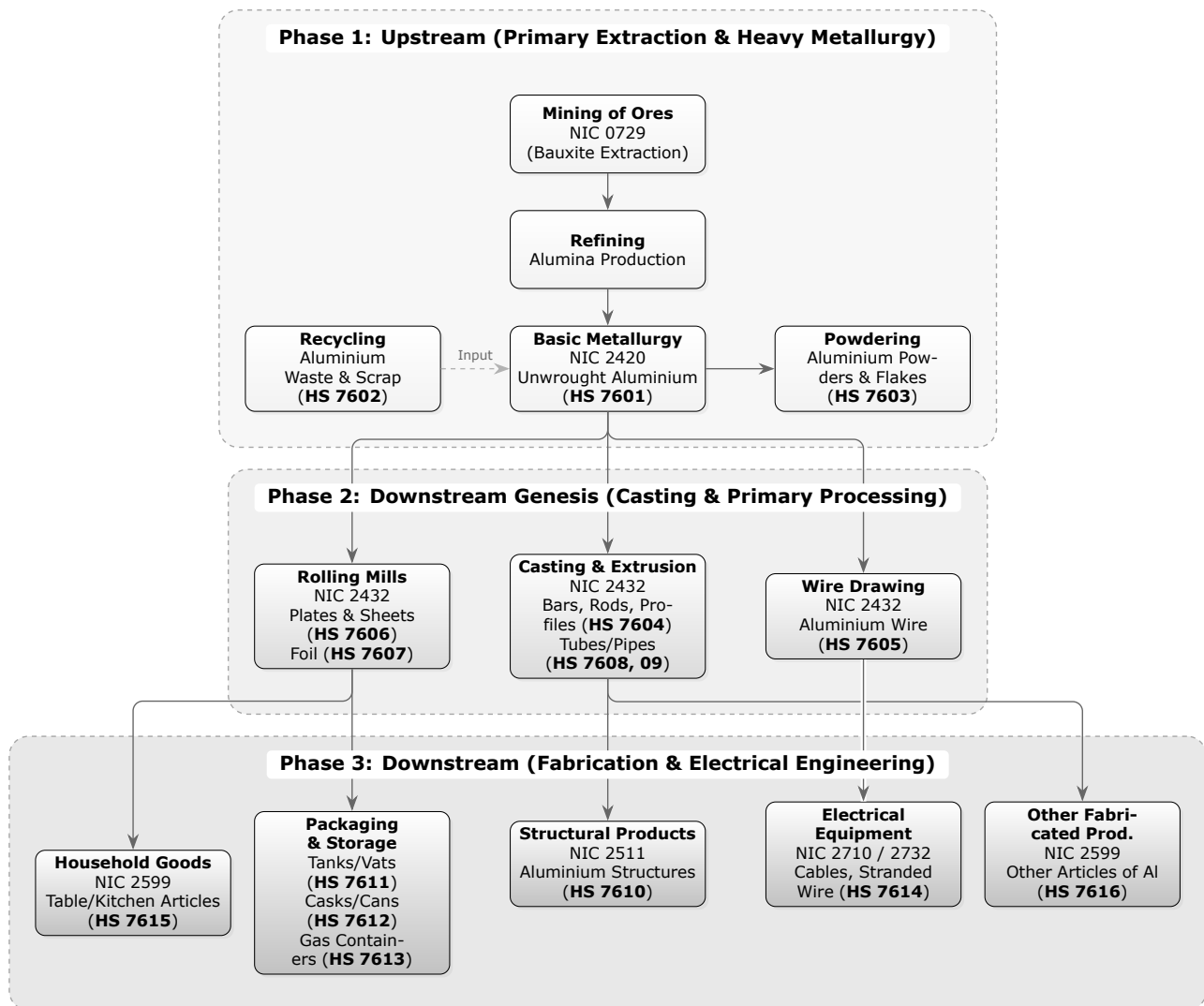


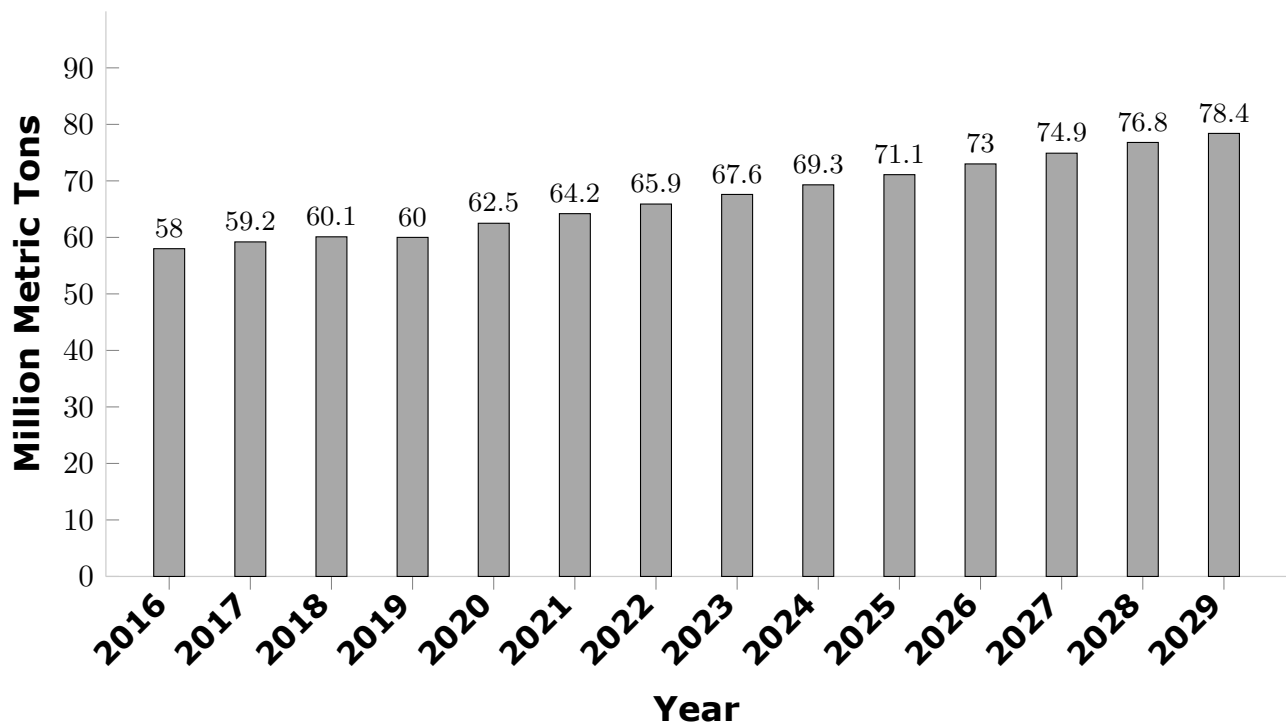
Figure 1.1: The Aluminium Value Chain, tracking industrial operations and HS Code progression.

2 International Trade Dynamics

The global aluminium landscape is characterized by steadily increasing consumption and shifting trade balances. Global primary aluminum consumption has grown consistently over the past decade, and this upward trajectory is reliably forecast to continue through 2029 (see Figure 2.1)¹⁷.

¹⁷Source: Statista.

Figure 2.1: Global Primary Aluminum Consumption and its Forecast (in million metric tons)

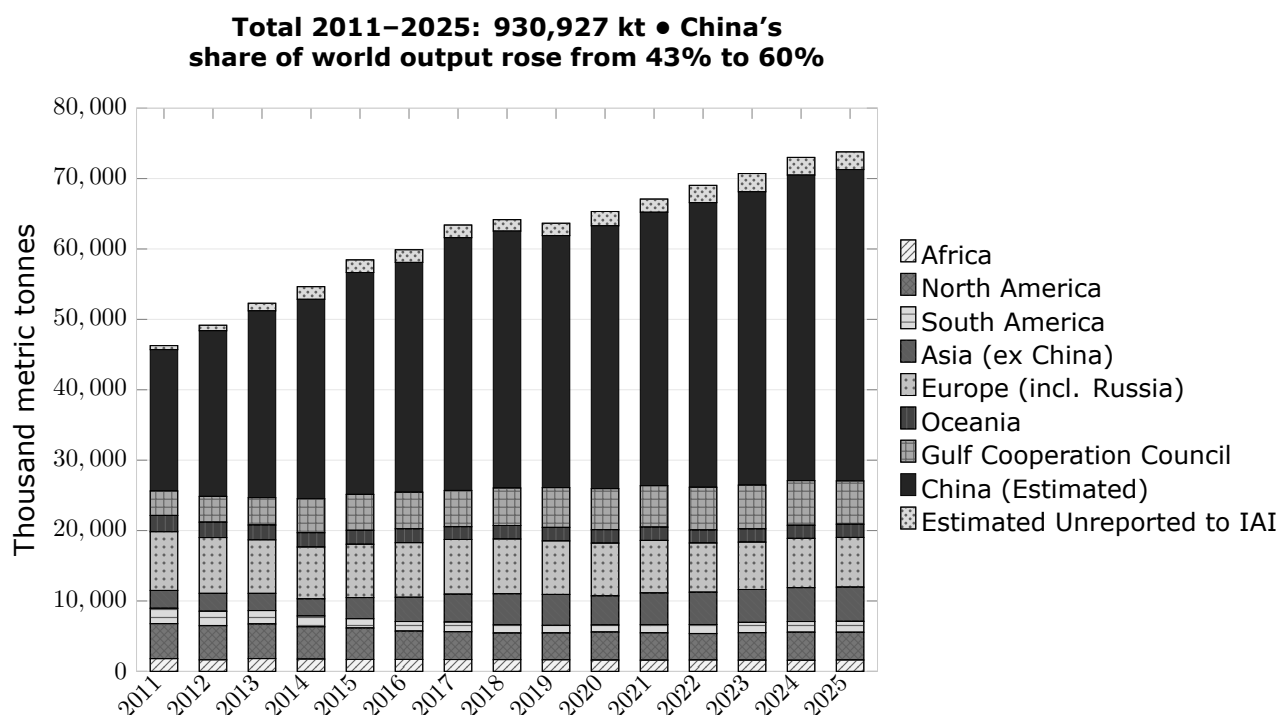


Source: Statista

Furthermore, global production output is heavily stratified by region, with massive historic output originating from Asia (particularly China) and the Gulf Cooperation Council (see Figure 2.2)¹⁸.

¹⁸Source: International Aluminium Institute.

Figure 2.2: Global Primary Aluminium Production by Region, 2011–2025 (thousand metric tonnes)

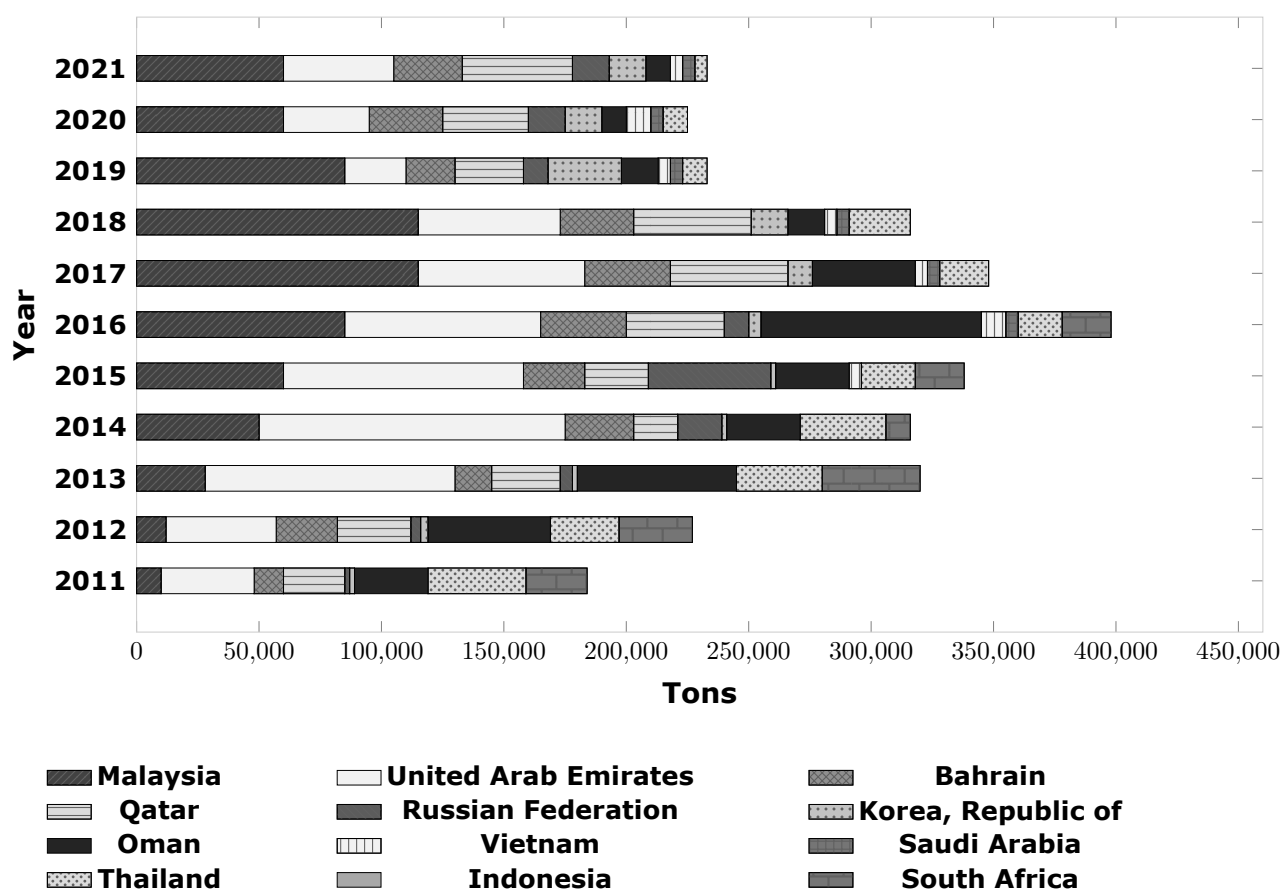


Source: International Aluminium Institute (IAI); annual totals of monthly reported primary aluminium production. Data for Jan–May 2026 are available but omitted for year-on-year comparability. From 2020 onward the IAI reports European output on a combined “Europe (incl. Russia)” basis; earlier “Western & Central” and “Russia & Eastern” figures are aggregated here for a consistent series.

Within India, the influx of primary aluminium is highly concentrated among a few key trading partners. An analysis of the top exporting countries of primary aluminum to India reveals that nations such as Malaysia, the United Arab Emirates, Qatar, and Bahrain thoroughly dominate the import volume (see Figure 2.3)¹⁹. This concentration clearly underscores the impact of Preferential Trade Agreements (PTAs) in the region and necessitates a critical review of the current tariff structures to protect domestic downstream manufacturers.

¹⁹Source: ITC Trade Map.

Figure 2.3: Top Exporting Countries of Primary Aluminum to India in 2021 (Tons)



Source: ITC Trade Map

2.1 Supply-Chain Fragility: Geopolitical Chokepoints, War-Risk Insurance, and Energy Costs

The heavy import concentration shown above is not just a matter of tariff arithmetic. It is a strategic vulnerability. Because the Inverted Duty Structure pushes domestic fabricators toward imported primary metal and scrap, it simultaneously exposes them to a chain of hidden, non-tariff costs that spike precisely when supply is most stressed. Roughly 60% of the Gulf Cooperation Council's 5.2 million tonnes of annual primary output is exported globally, and the primary maritime exit route is the Persian Gulf via the Strait of Hormuz—a chokepoint through which around 20% of the world's oil and LNG also transits²⁰. Given that Gulf producers—alongside Malaysia—dominate India's import basket, any disruption in this corridor transmits directly into the cost base of Indian downstream MSMEs.

Marine war-risk insurance is the sharpest of these hidden costs. Unlike standard hull or cargo cover, war-risk insurance in designated high-risk zones is priced dynamically by the Lloyd's Joint War Committee: underwriters reassess risk within 24-to-48-hour

²⁰"The Hidden Tax on Gulf Metal: How Marine War Premiums Are Rewriting the Economics of GCC Aluminium." *Discovery Alert* (2026). See also the synthesised trade-cost analysis in Ref. [15].

windows and issue rolling seven-day policies²¹. During periods of escalation the premium has climbed from a nominal fraction of hull value to as much as ten times its baseline, and these weekly surcharges pass immediately from shipowners to charterers and ultimately to aluminium buyers. Crucially, this dynamic decouples physical delivery costs from the London Metal Exchange (LME) headline: during recent Middle-East escalation, LME cash prices actually softened (retreating from roughly USD 3,680 to USD 3,480 per tonne as Chinese exports flooded warehouses) even as the Main Japanese Ports physical premium was projected to surge 27–30% to USD 450–460 per tonne²². For a price-taking Indian MSME, the practical consequence is that the landed cost of imported metal can climb sharply even while the “headline” aluminium price is falling.

Pandemic-style logistics shocks and direct energy exposure only compound this fragility. The COVID-19 disruption demonstrated how container shortages, port congestion, and freight-rate spikes can strand raw-material shipments and inflate regional premiums for months at a time—a systemic risk borne disproportionately by small fabricators without the balance sheet to hold buffer inventory or to self-insure. On the energy side, downstream remelting, casting, rolling, and extrusion are thermally intensive and typically fired by natural gas and LPG; fabricators therefore carry structural energy surcharges pegged to volatile gas indices, alongside diesel-linked fuel surcharges on the heavy, oversized freight that finished profiles and coils require²³. In aggregate, these logistics and energy liabilities mean the true cost of relying on imported primary aluminium is both higher and far more volatile than the nominal tariff suggests—strengthening, not weakening, the case for a domestic value chain insulated from these shocks.

²¹*Discovery Alert* (2026); see also *Comprehensive Economic Analysis of the Global Downstream Aluminium Industry* (2026), Ref. [15].

²²*Comprehensive Economic Analysis of the Global Downstream Aluminium Industry* (2026), Ref. [15]. The divergence illustrates that hedging LME exposure alone leaves the downstream buyer fully exposed to physical-premium volatility that cannot be hedged on the exchange.

²³*Comprehensive Economic Analysis of the Global Downstream Aluminium Industry* (2026), Ref. [15]. Fuel surcharges are quoted dynamically (e.g., per hundredweight against national diesel indices) and can swing materially month-to-month, while energy surcharges tier upward automatically once gas indices breach programmatic thresholds.

Table 2.1: Indian Downstream Sector: Primary vs. Secondary Aluminium Input Proportions

Fiscal Year	Primary (Unwrought) Share	Secondary (Scrap) Share
FY15	69%	31%
FY16	73%	27%
FY17	72%	28%
FY18	69%	31%
FY19	66%	34%
FY20	64%	36%
FY21	60%	40%
FY22	57%	43%
FY23	61%	39%
FY24	62%	38%

Note: Over the past ten fiscal years, the Indian downstream sector has demonstrated a clear pivot toward higher utilization of secondary aluminium. This data is corroborated by reports from CRISIL and the Ministry of Mines, based on a comprehensive review of domestic consumption data. The rising secondary share underscores the sector’s growing reliance on imported scrap (HS 7602)—cargo equally exposed to the marine war-risk and freight surcharges described above.

3 Effective Rate of Protection and the role of Inverted Duty Structure

3.1 The Current Tariff Regime, Import Volumes, and the Role of FTAs

The Indian downstream aluminium sector suffers from a severe Inverted Duty Structure (IDS), which is actively exacerbated by Free Trade Agreements (FTAs) and Preferential Trade Agreements (PTAs). India imported \$1.03 billion worth of unwrought primary aluminium (HS 7601) in 2024, totaling approximately 392,000 tons. Concurrently, India faces a massive influx of downstream imports, with China alone accounting for roughly 70% of downstream products (such as foils and plates) entering the country [4]²⁴. The composition of the two countries’ trade states the policy difference more plainly than any tariff schedule. Of India’s roughly US\$7 billion of aluminium exports in FY2025–26, **61.4 per cent was primary metal**; the corresponding share for China is **2.8 per cent**, value-added products making up the remaining 97.2 per cent [22]. In the same year India imported approximately US\$4.1 billion of *finished* aluminium goods—cables and conductors, packaging, engineering goods and automotive components—with about a quarter entering at low or zero duty under preferential agreements [22]. India exports the metal and imports the goods made from it.

The core of the IDS lies in the stark tariff differentials embedded within trade agreements. Primary aluminium (HS 7601) attracts a basic customs duty of 7.5% (effectively 8.25% with surcharges). However, finished goods from partners under the ASEAN-India Trade in Goods Agreement (AITIGA) and Comprehensive Economic Partnership Agreements (CEPA) with South Korea and the UAE often enter at 0% duty.

Crucially, these agreements are heavily skewed against Indian MSMEs. While India placed most downstream items in the “Normal Track” allowing duty-free access, partner nations have shielded their own markets. For instance, Malaysia imposes excessively

²⁴Ministry of Mines, Government of India. (2025). *Vision Document on Aluminium Metal for India*.

high Most Favoured Nation (MFN) duties (25% to 30%) on tariff lines between 7604 and 7608, and Indonesia places downstream products under “Sensitive Tracks” [4]²⁵. This creates a one-way street where foreign nations can import raw aluminium from India, add value, and dump finished goods back into the Indian market duty-free, completely crippling the competitiveness of domestic manufacturers.

Two domestic factors make the distortion worse still, by inflating the price of the raw input itself. First, because the primary smelting sector is highly consolidated—with a Herfindahl-Hirschman Index exceeding 3,100—upstream producers are able to price on an *import-parity* basis (IPP), benchmarking domestic prices to the LME plus the import duty and logistics premiums, alongside which a persistent spread of roughly Rs 30,000 to Rs 60,000 per tonne over international prices has been observed. The duty is one mechanism through which such a spread can be sustained, since it sets the floor beneath which imported metal cannot undercut domestic supply; we do not claim it is the sole driver, and a full decomposition against freight, insurance and physical premia lies outside this report’s scope²⁶. Second, statutory levies on energy and minerals—most notably the GST Compensation Cess on coal (escalated over the past decade to INR 400 per tonne) and, following the Supreme Court’s August 2024 ruling, retrospective state-level mineral cesses such as Odisha’s ORISED and Jharkhand’s MADA frameworks—inflate upstream production costs that are then passed down the chain via IPP²⁷. The combined effect is stark: Indian downstream capacity utilization has fallen to approximately 65% while consolidated upstream smelters run at close to 98%.

There is a further channel of cost volatility that tends to go unnoticed, and it sits inside the import-parity formula itself. Because domestic primary prices are constructed as the London Metal Exchange quotation plus a physical premium and a notional customs duty, and then converted into rupees at the prevailing US-dollar exchange rate, any depreciation of the rupee is mechanically passed through into the price paid by the downstream fabricator—even though the metal is produced entirely within India and the primary producer’s own costs, incurred in rupees, are unchanged. The inverted duty structure thus acquires a currency dimension: the import-parity model effectively transfers exchange-rate risk on *domestically produced* metal onto the downstream sector, so that a weaker rupee or a firmer LME raises input costs immediately, whereas a transparent, cost-based domestic benchmark would leave them unaffected. For thin-margin MSMEs, which earn in rupees and hold no natural dollar revenues against which to hedge, this currency-linked escalation compounds the margin compression already imposed by the duty inversion and the statutory cesses described above.

3.2 The Case for Countervailing and Anti-Dumping Measures against Subsidised Imports

The IDS does not operate in a vacuum; it is weaponised by trading partners—above all China—whose downstream exporters enjoy state support that Indian MSMEs cannot match. A 2019 study by the Organisation for Economic Co-operation and Development (OECD), reviewed in the European Commission’s Joint Research Centre assessment of the aluminium sector, found that government support pervades the entire aluminium

²⁵Ministry of Mines, *Vision Document on Aluminium Metal for India* (2025).

²⁶*Comprehensive Economic Analysis of the Global Downstream Aluminium Industry* (2026), Ref. [15]. The tariff, originally intended to protect domestic smelters, thus functions as an artificial pricing anchor that the downstream sector is forced to pay.

²⁷*Comprehensive Economic Analysis of the Global Downstream Aluminium Industry* (2026), Ref. [15]. Analysts estimate the mineral-cess ruling alone could raise landed raw-material costs by around 11% and compress downstream fabricator margins by 80–250 basis points, against a sector that already averages only ~5% margins.

value chain: each of the 17 largest global producers examined received financial or non-financial support, with five Chinese firms alone capturing over 85% of documented subsidies estimated at USD 20–70 billion across 2013–2017, largely in the form of energy subsidies and concessional finance²⁸.

What matters most for the Indian case is that China's distortion is deliberately built to favour its *downstream* sector at the direct expense of foreign fabricators. Beijing maintains an export tax of roughly 15% on primary aluminium and scrap while simultaneously granting a VAT rebate of 13–15% on exports of semi-finished and fabricated products (rolled products and extrusions)²⁹. The intent was demonstrated in May 2015, when China selectively eliminated its 15% export tax on certain semi-processed bars, rods, and strip while retaining the 30% export duty on most primary products; global producer share prices fell almost immediately, while the share price of China's largest state-owned producer rose by around 10%³⁰. This subsidy architecture, combined with structural overcapacity (Chinese primary output rose from 11% of world supply in 2000 to 56% in 2019, with idle smelting capacity exceeding 13 Mt), is precisely what floods the Indian market with below-cost foils and plates.

The right response to subsidised and dumped imports is not tariff rationalisation alone. It is the active use of the trade-remedy instruments the WTO framework already permits—anti-dumping duties on demonstrably below-cost imports and countervailing duties calibrated to neutralise the quantified foreign subsidy. India's downstream sector, unlike its trading partners, has been slow to seek such protection; a proactive regime of countervailing measures targeting subsidised Chinese and ASEAN semi-finished goods would restore competitive symmetry and complement the tariff reforms recommended in Section 5.

3.3 Empirical Evidence: The Erosion of Effective Protection, 2015–16 to 2024–25

The Effective Rate of Protection (ERP) provides the most appropriate empirical tool for analysing an Inverted Duty Structure. Whereas a nominal tariff records only the duty on the finished good, the ERP measures the protection afforded to the *value added* in a productive process, netting out the tariffs paid on that process's own intermediate inputs. Where input duties exceed output duties, the measure turns negative: the tariff regime taxes domestic transformation rather than protecting it, a paradox documented for Indian manufacturing more broadly by Pathania and Bhattacharjea [1]³¹.

Data and method

We estimate ERP for thirteen downstream product lines (HS 7604–7616) against six trading partners, at three points in time: 2015–16, 2019–20 and 2024–25. Estimates apply the modified Corden formulation set out in Appendix A, using Ministry of Statistics and Programme Implementation (MOSPI) Supply Use Tables for the input–output

²⁸Georgitzikis, K., Nita, V., Ciupagea, C., & Garbossa, E. (2021). *The aftermath of US tariffs on aluminium imports: An assessment of tariff impacts and a review of the EU aluminium industry*. EUR 30730 EN, Joint Research Centre, European Commission, Luxembourg. [5]; drawing on OECD (2019), *Measuring distortions in international markets: the aluminium value chain*. [6].

²⁹Georgitzikis et al. (2021), EU JRC, and OECD (2019). The export tax discourages the outflow of primary metal, making it artificially cheap to Chinese semis producers, while the VAT rebate subsidises the export of the value-added product—a textbook mechanism for capturing downstream value from competitors.

³⁰DeFrancesco, R. E., & Teslik, A. M. (2015). *China's Selective Elimination of Aluminum Export Duties Threatens Global Manufacturers*. Wiley Rein LLP Alert. [7].

³¹Pathania, K., & Bhattacharjea, A. (2020). *Inverted Duty Structures and the Paradox of Negative Effective Protection in India, 2000–2014*. Foreign Trade Review, 55(2), 139–167.

coefficients, and partner-specific applied (preferential where an agreement is in force, MFN otherwise) tariff schedules for output and import-weighted input duties. Looking at three widely spaced years rather than a single baseline lets us watch successive trade agreements take effect as they phase in.

Table 3.1: Effective Rate of Protection by downstream product line, 2024–25 (per cent)

Output Product	HS Code	Malaysia	S. Korea	Japan	Thailand	UAE	UK
Bars, Rods & Profiles	7604	−2.0	−2.0	−2.0	−2.0	+6.0	+38.0
Wires & Cables	7605	−1.0	−1.0	−1.0	−1.0	+7.0	+38.0
Flat Rolled Products	7606	−2.0	−2.0	−2.0	−2.0	+6.0	+37.0
Aluminium Foils	7607	−5.0	−5.0	−5.0	−5.0	+3.0	+34.0
Tubes and Pipes	7608	−1.0	−1.0	−1.0	−1.0	+9.0	+51.0
Tube/Pipe Fittings	7609	−3.0	−3.0	−3.0	−3.0	+8.0	+50.0
Aluminium Structures	7610	−2.0	−2.0	−2.0	−2.0	+9.0	+51.0
Reservoirs, Tanks, Vats	7611	−1.0	−1.0	−1.0	−1.0	+9.0	+51.0
Casks, Drums & Cans	7612	−3.0	−3.0	−3.0	−3.0	+8.0	+50.0
Containers for Gas	7613	−4.0	−4.0	−4.0	−4.0	+6.0	+48.0
Stranded Wire, Cables	7614	−1.0	−0.5	−0.5	−0.5	+10.0	+52.0
Utensils	7615	−6.0	−6.0	−6.0	−6.0	+15.0	+99.0
Castings	7616	−6.0	−6.0	−6.0	−6.0	+4.0	+46.0
Simple average		−2.9	−2.8	−2.8	−2.8	+7.7	+49.6

Note: Our estimates. A negative value denotes *disprotection*: the tariff structure leaves domestic value added lower than it would be under free trade. The UK column reflects MFN treatment during the estimation year and is reported as a non-preferential benchmark; the exceptionally high figure for utensils (HS 7615) reflects that line’s unusually low input-to-output value ratio and should be read with caution.

The result in Table 3.1 is strikingly uniform. In 2024–25, every single one of the fifty-two product–partner combinations involving Malaysia, South Korea, Japan and Thailand shows a negative ERP. This is not a quirk of one product line or one partner: it runs across the whole downstream range, from simple extrusions right through to finished utensils and castings.

The trend: protection erodes as agreements phase in

What the three estimation years reveal when set side by side is more telling still. Table 3.2 and Figure 3.1 summarise the comparison.

Table 3.2: Average ERP across thirteen downstream product lines, by partner and period (per cent)

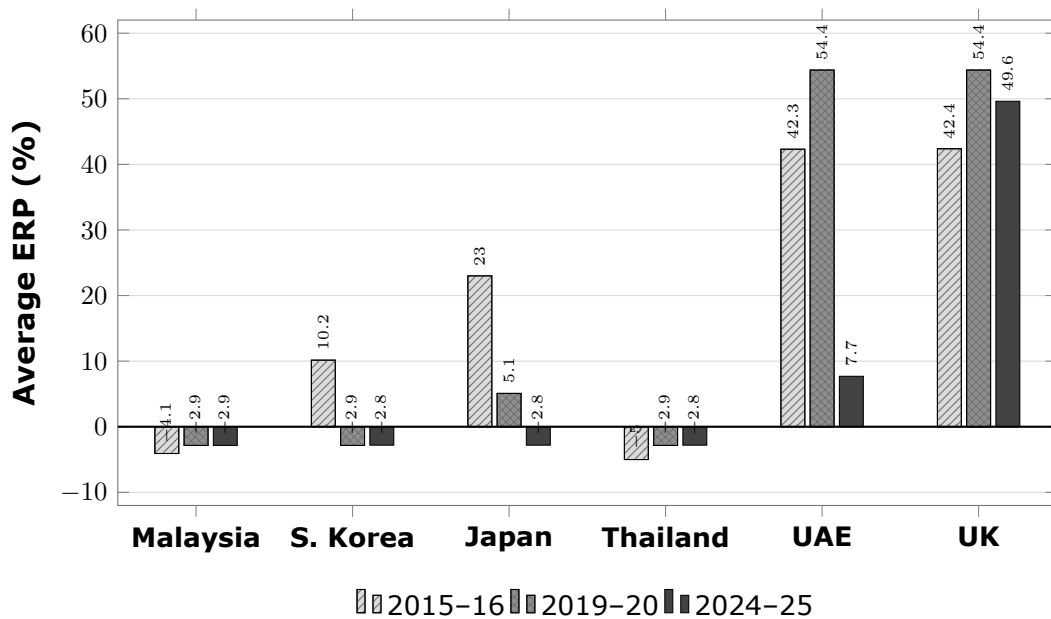
Partner	2015–16	2019–20	2024–25
Malaysia	–4.1	–2.9	–2.9
South Korea	+10.2	–2.9	–2.8
Japan	+23.0	+5.1	–2.8
Thailand	–5.0	–2.9	–2.8
UAE	+42.3	+54.4	+7.7
UK (non-preferential benchmark)	+42.4	+54.4	+49.6
<i>Product–partner cells with negative ERP, preferential partners only:</i>			
Malaysia, S. Korea, Japan, Thailand	26 of 52	34 of 52	52 of 52

Note: Simple (unweighted) averages of the thirteen product-line estimates. Our calculations.

Three things stand out here, and they matter more than any single-year snapshot could show.

- 1. Preferential partners have moved from positive to negative protection.** South Korea’s average ERP fell from +10.2% in 2015–16 to –2.9% by 2019–20; Japan’s fell from +23.0% to +5.1% and then to –2.8%. Malaysia and Thailand were already negative in 2015–16 and have remained so. The count of negative product–partner cells rose from 26 of 52, to 34 of 52, to all 52.
- 2. The UAE illustrates the mechanism directly.** Effective protection against UAE-origin goods stood at +42.3% in 2015–16 and +54.4% in 2019–20. Following the India–UAE Comprehensive Economic Partnership Agreement, it collapsed to +7.7% in 2024–25—a fall of roughly forty-seven percentage points, with no corresponding reduction in the duty payable on the sector’s primary input.
- 3. A non-preferential benchmark holds the method constant.** The United Kingdom, trading on MFN terms across these years, retained an average ERP of +49.6% in 2024–25. It is computed on the same thirteen product lines with the same estimator as the preferential partners and differs only in that its goods enter at most-favoured-nation rates; it is a benchmark rather than a control, and no causal weight is placed on the comparison beyond that. The divergence between this benchmark and the preferential columns isolates the effect of the agreements themselves rather than of any general change in India’s tariff schedule.

Figure 3.1: Average Effective Rate of Protection by partner and period



Source: Our estimates from MOSPI Supply Use Tables and partner tariff schedules. Bars below the zero line denote negative effective protection. The UK series is a non-preferential (MFN) benchmark.

Read together, these estimates point to something more troubling than a one-off distortion. The disprotection facing India's downstream aluminium sector is not standing still—it is widening, and it widens in step with the order in which India's trade agreements have come into force.

3.4 The "Lose-Lose" Scenario for the Indian Consumer

As established in trade literature, increased access to imported intermediate varieties is a primary driver of domestic product scope expansion and innovation [2]³². By maintaining high input tariffs, the current policy stifles domestic product diversification. This creates a "lose-lose" scenario for the Indian consumer:

1. **Vulnerable Supply Chains:** Consumers become overly reliant on imported finished goods from PTA nations and China, reducing domestic supply chain resilience.
2. **Stifled Innovation:** High costs for intermediate inputs deprive consumers of new, customized domestic product varieties.
3. **Eroded Purchasing Power:** The destruction of the labour-intensive MSME downstream sector results in widespread job losses, directly impacting the economic well-being of the Indian working class.

³²Goldberg, P. K., Khandelwal, A. K., Pavcnik, N., & Topalova, P. (2010). *Imported Intermediate Inputs and Domestic Product Growth: Evidence from India*. The Quarterly Journal of Economics, 125(4), 1727–1767.

4 Policy Plight of MSMEs in the Aluminium Industry

4.1 Measuring the Sector: Where the Value Chain Divides

The claims made so far deserve to be tested against official statistics rather than industry assertion, and for that we use the **Annual Survey of Industries (ASI)** [10]. Everything then turns on where the chain is divided, and the division used here is not a matter of taste: it is the one the theory behind Section 3 requires.

Corden [16] defines the object of effective-protection analysis as an *activity*—a stage of conversion from traded input to traded output, whose protection depends jointly on the tariff on that output and the tariffs on its inputs. The aluminium chain therefore divides where metal stops being something a producer makes and becomes something a producer buys. On one side stands the metal itself, **HS 7601–7603**, carrying the basic customs duty of 7.5 per cent. On the other stand all the activities that buy it at a duty-inclusive price and convert it—extrusion, rolling, wire drawing, foil, shape casting, structural fabrication, utensil manufacture. Each faces the same wedge between a protected input and an output whose own tariff has been bargained away, so on Corden’s criterion they are one class of activity: precisely the thirteen lines **HS 7604–7616** on which Section 3 estimates effective protection.

The pathology has a precedent. Balassa found the effective rate on United States wool clothing turning *negative* because wool yarn and fabric were themselves protected [17], and established tariff escalation as the systematic feature that penalises processing [18]; Pathania and Bhattacharjea document the same result across Indian manufacturing [1]. Industry practice draws the line in the same place: the OECD treats semi-fabrication as a *downstream* activity and finds upstream support conferring advantage on it [6]. The material-flow literature is finer-grained—Cullen and Allwood separate semi-fabrication from fabrication as distinct process steps [19, 20]—but a tariff does not distinguish a rolling mill from a window fabricator, and where the distinction might matter it is tested rather than assumed (Appendix B).

Applying the rule yields the concordance in Table 4.1 [11].

Table 4.1: The thirteen ERP product lines mapped to their ASI manufacturing classes

HS line	Product	Converting activity	NIC class
<i>Upstream — the dutiable input, not part of the ERP basket</i>			
7601	Unwrought aluminium	Smelting from alumina	2420
7602	Waste and scrap	Secondary refining	2420
7603	Powders and flakes	Atomising, milling	2420
<i>Downstream — the thirteen ERP basket lines</i>			
7604	Bars, rods, profiles	Extrusion	2432
7605	Wire	Wire drawing	2432
7606	Plates, sheets, strip	Flat rolling	2432
7607	Foil	Foil rolling	2432
7608	Tubes and pipes	Extrusion, drawing	2432
7609	Tube and pipe fittings	Fabrication, machining	2599
7610	Structures, doors, windows, façades	Structural fabrication	2511
7611	Reservoirs, tanks, vats	Tank fabrication	2512
7612	Casks, drums, cans	Container fabrication	2599
7613	Containers for compressed gas	Cylinder fabrication	2512
7614	Stranded wire, cables, plaited bands	Stranding, plaiting	2599
7615	Table, kitchen and household articles	Utensil manufacture	2599
7616	Castings and other articles	Shape casting, general fabrication	2432, 2599

Notes. Upstream = NIC 2420; downstream = NIC 2432 + 2511 + 2599. NIC 0729 (bauxite) and NIC 2512 are not separately available in the ASI All-India summary. NIC 2710, 2732, 2733 and 3030 are excluded: their principal outputs fall in HS Chapters 85 and 88, which on Corden's criterion makes them a different activity, protected by a different set of tariffs. Appendix B states the rule in full, together with the treatment of classes that are not exclusive to aluminium and the effect of every alternative tested.

Three things should be said before the evidence, each developed in Appendix B. All values are nominal, at current prices. All stated statistics run to **2022–23**. And the comparison is biased *against* our own conclusion: semi-fabrication carried out inside integrated smelters is recorded upstream, NIC 2512 is not separately published, and the ASI frame excludes the unincorporated units in which downstream fabrication is concentrated. All three inflate the measured upstream segment and deflate the downstream one, so every gap below is a lower bound.

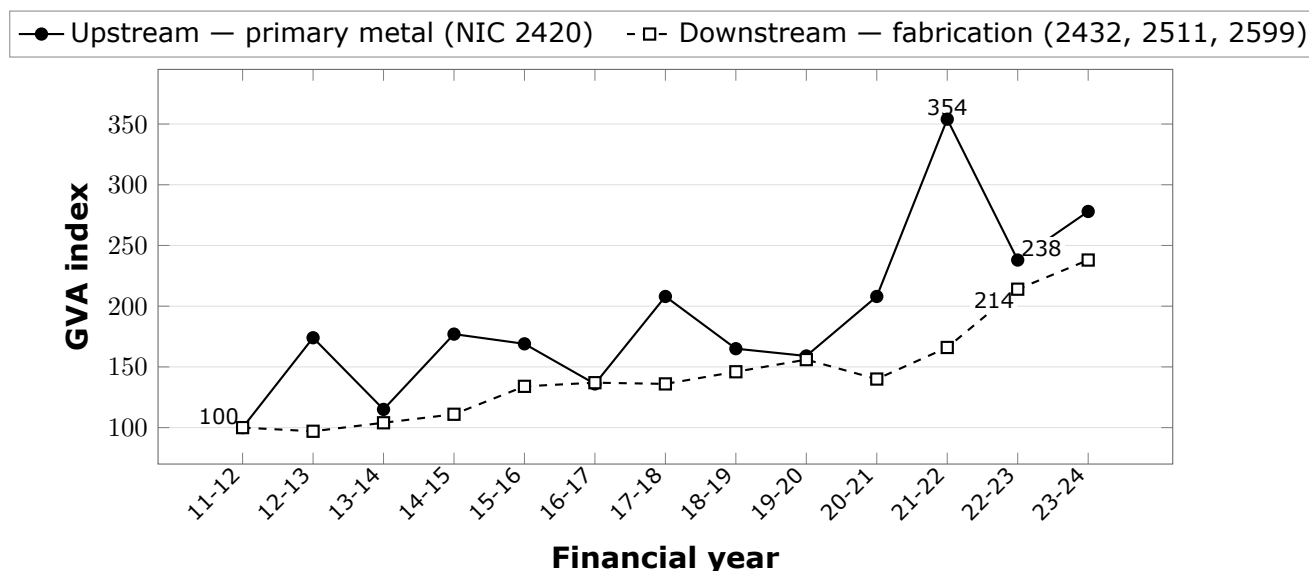
What the evidence is, and is not. What follows is association, not an identified causal effect. The data show that the class the effective-protection analysis identifies as most disprotected is also the class whose value added, employment and tax contribution have stagnated, and they supply a mechanism—an input-cost share of 83.1 per cent—by which the first could produce the second. There is no counterfactual India without the duty, the effective-protection series runs to three years against twelve of ASI data, and the construction cycle is not separately controlled for. What can be said is that the association holds in sign in every year of the record and is concentrated in precisely the class the tariff analysis singles out. A causal claim would require firm-level data spanning a tariff change; we do not make one.

4.2 Value Added and Employment

Over the eleven years to 2022–23 the two segments did not grow alike. Upstream gross value added rose **138.4 per cent** against **114.5 per cent** downstream, leaving them of comparable size—INR 42,662 crore against INR 41,653 crore—where downstream had begun the period the larger (Figure 4.1). Measured between three-year averages at

each end, which removes the sensitivity to a volatile base year, the gap is similar: 105.6 against 73.0 per cent, compound rates of 8.34 and 6.28. Over precisely the years in which effective protection for the thirteen downstream lines was eroding, the protected node expanded faster and overtook the disprotected one.

Figure 4.1: Gross value added, index 2011–12 = 100



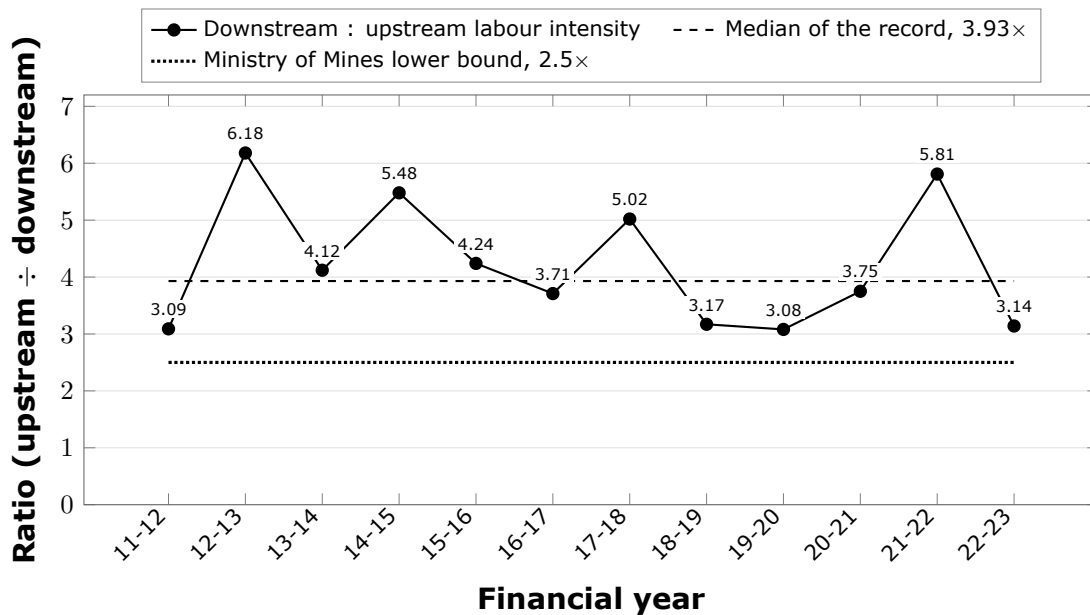
Source: our calculations from ASI All-India summary results.

The employment arithmetic runs the other way, and it is the heart of the matter. That upstream value added is generated by 170,871 persons engaged; the downstream classes engage **523,899**. Per head, upstream value added stood at INR 24.97 lakh in 2022–23 against INR 7.95 lakh downstream.

4.3 Labour Intensity: The Central Finding

Each rupee of value added created downstream therefore sustains **3.14 times** more employment than the same rupee created in primary metal. Because this ratio carries most of the section's weight, Figure 4.2 shows it for every year rather than for the single year quoted.

Figure 4.2: Labour intensity of downstream fabrication relative to primary metal



Source: our calculations from ASI All-India summary results.

The ratio never falls below **3.08** and reaches 6.18; its median is **3.93**. In every one of the twelve years downstream is the more labour-intensive segment, and in every one the ratio exceeds 2.5—the lower bound of the Ministry of Mines’ own assessment that downstream activity generates two-and-a-half to three times the employment of primary production [4]. Our estimate corroborates the Ministry’s from an independent source and sits at the upper end of its range. The peaks are metal-price years, which raise upstream value added without raising upstream employment; the ratio is therefore lowest in the years least flattering to the argument, and 2022–23, the year quoted throughout, is one of them.

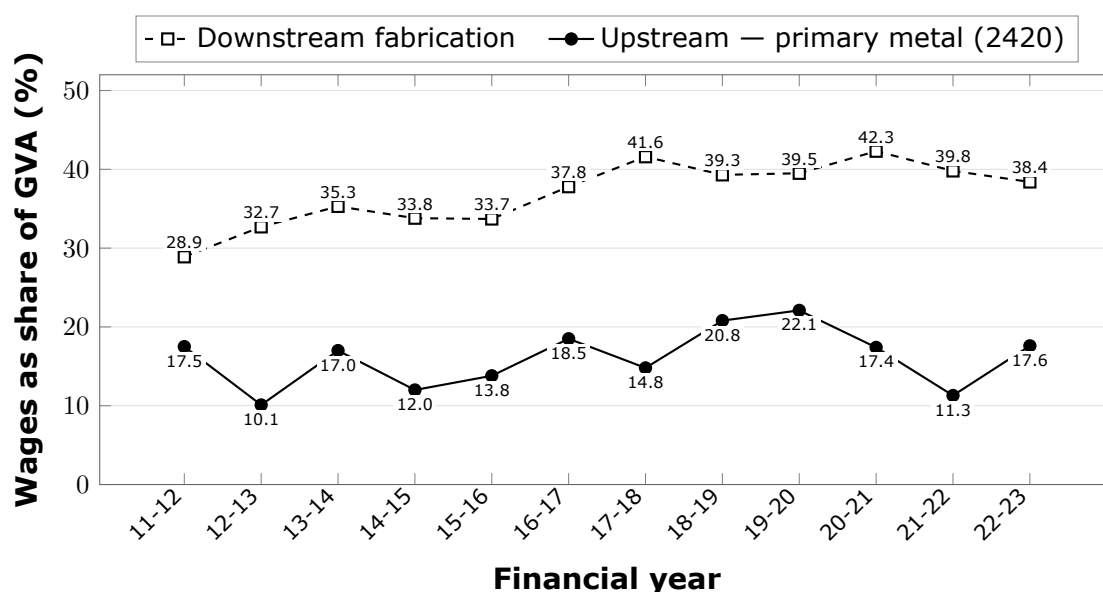
The same finding, per tonne of metal. Value added is the right denominator for an economic comparison, but a policymaker deciding where a tonne of metal should be converted will ask a blunter question. On that measure the answer is **41 persons engaged per thousand tonnes upstream against 210 to 262 downstream**, a ratio of **5.1 to 6.4**. The two normalisations are related exactly, so the segment that is more labour-intensive per rupee is also, and more strongly, more labour-intensive per tonne.³³

4.4 Labour Share: Where the Value Added Goes

Employment counts measure how many people a segment engages, not how much of the value created reaches them. On that second question the divergence is starker, and it depends on no assumption about how persons engaged are counted. In 2022–23 downstream paid out **38.4 per cent** of gross value added as wages and salaries against **17.6 per cent** upstream (Figure 4.3).

³³Derived in Appendix B. Tonnage is not an ASI variable, so this statistic supports the value-added measure rather than replacing it.

Figure 4.3: Wages and salaries as a share of gross value added (per cent)



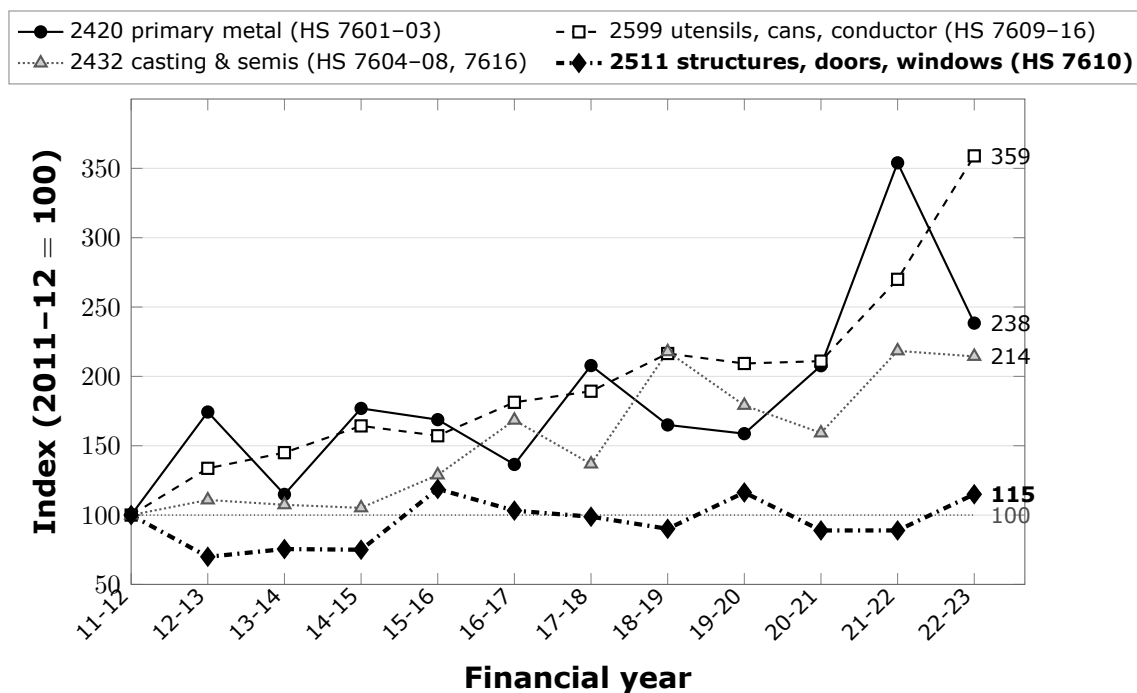
Source: our calculations from ASI All-India summary results.

The gap is not a single-year artefact: the two series never cross, and the downstream share has *risen* over the period while upstream has oscillated with no trend, its troughs coinciding with the metal-price peaks. A rupee of value added retained downstream delivers roughly **2.2 times as much to household income** as the same rupee retained upstream—the cleanest single statistic in this section, and the one least exposed to methodological objection.

4.5 Where the Pressure Falls: Structural Products

The downstream aggregate conceals a divergence that matters for policy, because it maps onto the effective-protection estimates directly (Figure 4.4).

Figure 4.4: Gross value added by class, index 2011–12 = 100



Source: our calculations from ASI All-India summary results.

Structural metal products (NIC 2511)—the class that makes HS 7610 aluminium frameworks, doors, windows and façades—grew gross value added by **15.0 per cent** over eleven nominal years, and by 19.4 per cent on the three-year-average basis. Against any plausible measure of cumulative inflation that is a deep real contraction. Its persons engaged rose 0.2 per cent and *fell* 5.9 per cent between the three-year endpoints; its contract workforce fell 1.1 per cent; its estimated tax contribution declined 16.1 per cent. Every indicator points the same way.

This is not an arbitrary class to find under pressure. HS 7610 carries among the highest disprotection in the entire ERP table—by 2024–25, negative effective protection against Malaysia, Korea, Japan and Thailand alike, against a non-preferential benchmark of +0.51. The single ASI class most exposed to duty-free finished imports is the single ASI class that has not grown.

Table 4.2: Percentage change by NIC class, 2011–12 to 2022–23 (nominal)

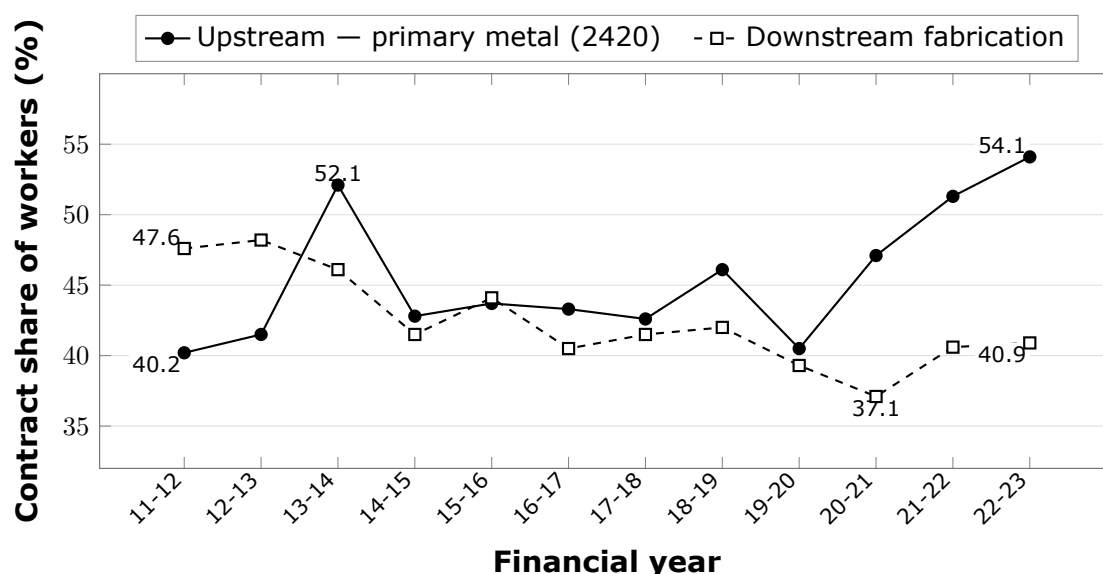
NIC class	HS lines	GVA	Persons	Workers	Contract	Wages	Tax
<i>Upstream — the protected input</i>							
2420 Primary metal and semis	7601–7603	+138.4	+39.5	+48.7	+100.0	+139.6	+181.6
<i>Downstream — the ERP basket</i>							
2432 Casting of non-ferrous metals	7604–08, 7616	+114.4	+11.9	+14.8	+28.6	+153.6	+196.8
2511 Structural metal products	7610	+15.0	+0.2	+0.3	–1.1	+102.6	–16.1
2599 Other fabricated metal products	7609, 7612–16	+259.0	+50.9	+49.8	+16.0	+263.6	+412.4
Upstream total		+138.4	+39.5	+48.7	+100.0	+139.6	+181.6
Downstream total		+114.5	+27.4	+27.1	+9.0	+184.5	+110.1

Source: our calculations from ASI All-India summary results.

4.6 Employment Composition

Chaurey [3] shows that Indian manufacturers use contract labour as their margin of flexibility, because permanent workers are difficult to adjust under employment protection legislation. The two segments have used that margin in opposite directions (Figure 4.5). Upstream, the contract workforce **doubled**, from 37,228 to 74,449, and its share of total workers climbed from 40.2 to **54.1 per cent**. Downstream, contract employment rose 9.0 per cent and the share *fell*, from 47.6 to **40.9 per cent**.

Figure 4.5: Workers employed through contractors as a share of total workers (per cent)



Source: our calculations from ASI All-India summary results.

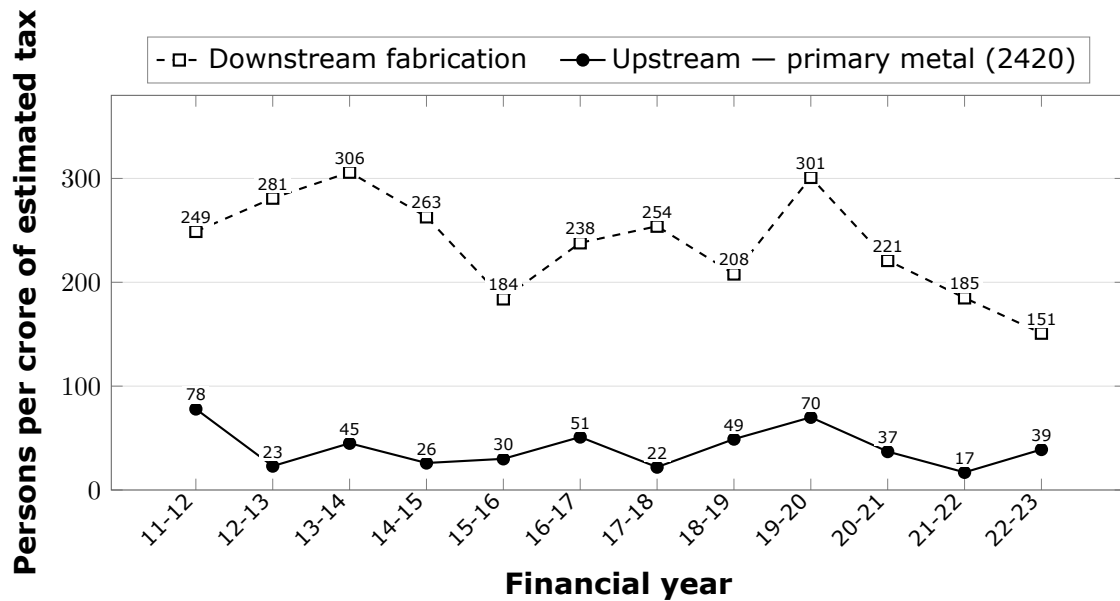
The falling downstream share is not improving job quality; it is the signature of a segment that has stopped hiring. Downstream nominal value added more than doubled while its flexible workforce grew nine per cent—the segment is running existing capacity harder rather than adding to it. Upstream, by contrast, expanded employment and casualised it. A segment under sustained margin pressure does not, on this evidence, shed workers so much as stop taking them on, and foregone employment appears in no redundancy statistic.

4.7 Corporate Tax: Employment Yield and Revenue Stability

The ASI does not publish tax paid, so corporate tax here is an *estimate* of the tax base attributable to each segment rather than a record of receipts. On levels, upstream is ahead: an estimated INR 4,411 crore in 2022–23 against INR 3,473 crore downstream. Levels are the wrong test for a tariff decision, for two reasons.

First, the employment attached to each rupee differs by nearly a factor of four. The downstream classes support **151 persons engaged per crore** of estimated corporate tax against **39** upstream, and the ordering holds in all twelve years without the gap ever falling below a factor of two (Figure 4.6).

Figure 4.6: Persons engaged per INR crore of estimated corporate tax



Source: our calculations from ASI All-India summary results.

Second, the two revenue streams are of different quality. Upstream tax has run from INR 1,567 crore to INR 8,996 crore at the 2021–22 price peak and back to INR 4,411 crore a year later; downstream tax has risen in a near-monotone line from INR 1,653 crore to INR 3,473 crore. The underlying volatility runs the same way: the standard deviation of year-on-year log growth in value added is **0.373** upstream against **0.105** downstream, so the protected node's value added is **3.6 times as volatile**. Rationalising the duty on primary aluminium therefore trades a *cyclical* revenue stream, thinly attached to employment, for relief to a segment on which roughly four times as many livelihoods depend per rupee of tax at stake.

The transmission mechanism, and its magnitude. Bought-in inputs account for **83.1 per cent** of downstream output value—four-fifths of every rupee of turnover is purchased material, overwhelmingly metal bought at import-parity prices set by a duty the fabricator cannot avoid and, on the evidence of Section 3, cannot pass into an output price that competes with duty-free finished imports. Because value added is what remains after inputs, the ratio of the two—**4.91**—is the factor by which any proportional change in input cost is amplified by the time it reaches value added. Where metal is 60 per cent of the downstream input bill, a one per cent rise in the metal price removes about **3 per cent** of downstream value added and the 7.5 per cent duty something over a fifth of it; upstream the figure is nil, because a smelter does not buy the metal it makes.³⁴ This is Corden's arithmetic in its plainest form [16]: an input-price wedge of that size cannot be a marginal factor in downstream margins.

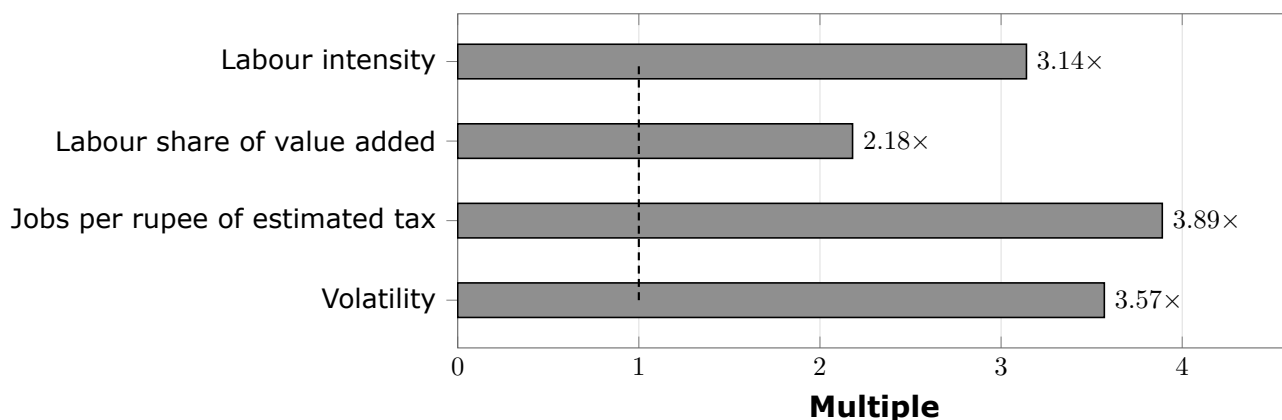
4.8 The Two Segments at a Glance

Figure 4.7 collects the four measures on which this section rests. Each is expressed as a multiple so the four are comparable, and each is drawn from a different part of the

³⁴Derivation and the full range in Appendix B.

ASI schedule—value added and employment, wages, corporate tax, and the variance of value added. They are not four versions of one statistic.

Figure 4.7: How far the two segments differ, 2022–23 (parity = 1.0×)



Source: our calculations from ASI All-India summary results.

The first three favour downstream on employment and household-income grounds; the fourth favours it on revenue stability, since it is *upstream* value added that is the more volatile. Every claim above was tested against the obvious objections—choice of year and base year, the exclusion of any single class, the placement of semi-fabrication, the width of the downstream basket, and the possibility that the classes should be scaled by their aluminium content. None changes a sign, and the largest moves the labour-intensity ratio only from 3.14 to 2.61. The tests and the assumptions they probe are in Appendix B.

4.9 Compliance Headwinds: CBAM, ESG Norms, and the CAPEX Barrier to Exports

Even as domestic MSMEs are squeezed at home, their export potential is increasingly constrained by environmental-compliance regimes in destination markets. From January 2026 the European Union's Carbon Border Adjustment Mechanism (CBAM) entered its definitive phase, converting the embedded-carbon content of imported aluminium into a direct financial obligation: importers must surrender certificates equivalent to the EU Emissions Trading System carbon price, and exporters must supply installation-specific, meter-based, third-party-verified emissions data rather than industry averages³⁵. Where a producer cannot furnish verified data, the customer is assigned punitive region-specific default values calculated from the worst-emitting 10% of installations—penalising precisely the small Indian fabricators least able to afford sophisticated carbon accounting. A parallel UK CBAM is scheduled from January 2027, and similar ESG-driven disclosure and due-diligence norms are proliferating across OECD markets.

At bottom, this is a capital problem. Because a large share of an aluminium product's embedded carbon is inherited from upstream coal-fired Indian smelting, meaningful decarbonisation requires cost-intensive technology upgrades—greenfield low-emission smelting routes can entail capital expenditure on the order of USD 350 per tonne, more than double conventional processes, and comprehensive sectoral decarbonisation implies

³⁵"How to Calculate Embedded Emissions for UK Aluminium Manufacturers." *Below280 / Decerna* (2026); European Commission CBAM Regulation (EU) 2023/956. [8].

multi-billion-dollar investment³⁶. For a fragmented downstream sector already operating on ~5% margins and paying import-parity prices for its metal, the CAPEX required to meet CBAM and ESG thresholds is prohibitive. Unless input costs are rationalised and targeted green-transition finance is extended, these compliance regimes will function as a de facto non-tariff barrier that locks Indian downstream products out of high-value export markets—compounding the domestic damage inflicted by the Inverted Duty Structure.

5 Conclusion

To resolve the structural imbalances within India's aluminium ecosystem and safeguard its strategic future, we propose the following targeted policy interventions [4]³⁷:

- **Tariff Rationalization on Primary Inputs:** The government must urgently address the Inverted Duty Structure by rationalizing the 7.5% basic customs duty on unwrought primary aluminium (HS 7601) and associated raw materials. Aligning the duties on these essential inputs with the zero-duty or low-duty finished goods entering under FTAs is necessary to restore a level playing field for domestic secondary manufacturers and MSMEs.
- **Deploy Trade-Remedy Instruments against Subsidised Imports:** Beyond tariff realignment, India should proactively invoke WTO-consistent anti-dumping and countervailing duties on demonstrably subsidised and below-cost semi-finished imports—particularly from China, whose 13–15% VAT rebates and export-tax architecture directly subsidise its downstream exporters. Trade remedies calibrated to the quantified foreign subsidy would neutralise the distortion at source.
- **Renegotiation of Free Trade Agreements (FTAs):** Trade agreements with ASEAN nations (specifically Malaysia and Indonesia) must be revisited to address the non-reciprocal tariff barriers. Downstream items currently placed on Sensitive Tracks or subjected to high MFN duties by partner nations must be renegotiated to allow fair market access for Indian MSME exports.
- **Establish a Transparent Domestic Pricing Framework:** Currently, primary producers operate on an Import Parity Pricing (IPP) model tied to the London Metal Exchange (LME). This effectively inflates domestic prices despite India's inherent low-cost production advantages. Shifting toward a transparent, cost-based domestic pricing framework will shield downstream MSMEs from global market volatility and severe margin pressures.
- **Enable Green-Transition Finance for CBAM/ESG Compliance:** To preserve export access under the EU and UK CBAM and proliferating ESG norms, policymakers should pair input-cost relief with targeted concessional finance and incentives for the high-CAPEX technology upgrades required to produce verified low-carbon aluminium, ensuring MSMEs are not locked out of high-value markets.
- **Streamline Regulatory Burdens and QCOs:** Non-tariff barriers, particularly Quality Control Orders (QCOs) on upstream product categories, restrict access to necessary imported raw materials and impose heavy compliance costs on MSMEs.

³⁶Ministry of Mines / NITI Aayog, *Roadmap for Aluminium Sector Decarbonisation*. [9]. The high upfront CAPEX and elevated cost of capital place verified low-carbon compliance out of reach for most MSMEs absent policy support.

³⁷Ministry of Mines, Government of India. (2025). *Vision Document on Aluminium Metal for India*.

Policymakers must continue to limit QCO burdens on critical intermediate inputs to prevent supply chain disruptions in the downstream sector.

Correcting the tariff anomalies and market rigidities facing the downstream aluminium industry is a strategic economic imperative. As India navigates the pressures of global supply chains, geopolitical chokepoints, subsidised foreign competition, rapid urbanization, and the impending EU CBAM, empowering the secondary manufacturing sector is essential. The downstream sector has the capacity to generate nearly triple the employment of the primary sector. By rationalizing input tariffs, deploying trade remedies against subsidised imports, renegotiating unbalanced FTAs, ensuring transparent domestic pricing, enabling green-transition finance, and easing regulatory burdens, policymakers can unlock massive GVA potential. Crucially, these reforms will safeguard the thousands of MSMEs that form the backbone of this industry, thereby protecting and expanding a vital, pro-employer base for India's growing workforce.

References

- [1] Pathania, K., & Bhattacharjea, A. (2020). *Inverted Duty Structures and the Paradox of Negative Effective Protection in India, 2000-2014*. *Foreign Trade Review*, 55(2), 139–167.
- [2] Goldberg, P. K., Khandelwal, A. K., Pavcnik, N., & Topalova, P. (2010). *Imported Intermediate Inputs and Domestic Product Growth: Evidence from India*. *The Quarterly Journal of Economics*, 125(4), 1727–1767.
- [3] Chaurey, R. (2015). *Labor regulations and contract labor use: Evidence from Indian firms*. *Journal of Development Economics*, 114, 224–232.
- [4] Ministry of Mines, Government of India. (2025). *Vision Document on Aluminium Metal for India*. New Delhi, India.
- [5] Georgitzikis, K., Nita, V., Ciupagea, C., & Garbossa, E. (2021). *The aftermath of US tariffs on aluminium imports: An assessment of tariff impacts and a review of the EU aluminium industry*. EUR 30730 EN, Joint Research Centre, European Commission, Luxembourg. doi:10.2760/46784.
- [6] Organisation for Economic Co-operation and Development. (2019). *Measuring distortions in international markets: the aluminium value chain*. OECD Trade Policy Papers, No. 218, OECD Publishing, Paris.
- [7] DeFrancesco, R. E., & Teslik, A. M. (2015). *China's Selective Elimination of Aluminium Export Duties Threatens Global Manufacturers*. Wiley Rein LLP International Trade Alert, May 22, 2015.
- [8] European Commission. (2023). *Regulation (EU) 2023/956 establishing a Carbon Border Adjustment Mechanism (CBAM)*. Official Journal of the European Union; definitive phase from January 2026.
- [9] Ministry of Mines / NITI Aayog, Government of India. *Roadmap for Aluminium Sector Decarbonisation*. New Delhi, India.
- [10] National Statistical Office, Ministry of Statistics and Programme Implementation, Government of India. *Annual Survey of Industries (ASI), All-India summary results, 2011–12 to 2023–24*. New Delhi. Variable definitions per the ASI schedule and published concepts and definitions.
- [11] Central Statistical Organisation, Ministry of Statistics and Programme Implementation, Government of India. (2008). *National Industrial Classification (NIC) 2008*. New Delhi.
- [12] AL Circle. (2026). *Vedanta Aluminium to rally 22%? Nuvama initiates 'Buy' with 29% EBITDA forecast*. 14 July 2026. <https://www.alcircle.com/news/vedanta-aluminium-to-rally-22-nuvama-initiates-buy-with-29-ebitda-forecast-120324>
- [13] Arora, A. (2023). *Assessing the Impact of CBAM and Tariffs in the Indian Aluminum Industry*. IIFT Working Paper Series, WPS No. EC-23-65, Indian Institute of Foreign Trade, New Delhi.
- [14] Ben-Salha, O., Zmami, M., Waked, S. S., Najjar, F., & Alenazi, Y. M. (2025). *On the time-varying spillover between nonferrous metals prices, geopolitical risks, and global economic policy uncertainty*. *Economic Change and Restructuring*, 58(5).

- [15] *Comprehensive Economic Analysis of the Global Downstream Aluminium Industry: Regulatory Frameworks, Cost Modeling, and Supply Chain Mechanics*. Synthesis of trade data and peer-reviewed literature, 2026.
- [16] Corden, W. M. (1966). *The Structure of a Tariff System and the Effective Protective Rate*. *Journal of Political Economy*, 74(3), 221–237.
- [17] Balassa, B. (1965). *Tariff Protection in Industrial Countries: An Evaluation*. *Journal of Political Economy*, 73(6), 573–594.
- [18] Balassa, B. (1968). *Tariff Protection in Industrial Nations and Its Effects on the Exports of Processed Goods from Developing Countries*. *Canadian Journal of Economics*, 1(3), 583–594.
- [19] Cullen, J. M., & Allwood, J. M. (2013). *Mapping the Global Flow of Aluminum: From Liquid Aluminum to End-Use Goods*. *Environmental Science & Technology*, 47(7), 3057–3064. doi:10.1021/es304256s.
- [20] Bertram, M., Martchek, K. J., & Rombach, G. (2009). *Material Flow Analysis in the Aluminum Industry*. *Journal of Industrial Ecology*, 13(5). doi:10.1111/j.1530-9290.2009.00158.x.
- [21] Bohlin, L., & Widell, L. M. (2006). *Estimation of commodity-by-commodity input-output matrices*. *Economic Systems Research*, 18(2), 205–215. doi:10.1080/09535310600653164.
- [22] Global Trade Research Initiative (2026). *India's aluminium trade: primary metal exports and finished-product imports, FY2025–26*. GTRI Policy Report, June 2026; figures as reported in *Business Standard*, 18 June 2026.
- [23] Miller, R. E., & Blair, P. D. (2009). *Input-Output Analysis: Foundations and Extensions*, 2nd ed. Cambridge University Press, ch. 5.
- [24] Ministry of Finance, Government of India (2026). *Union Budget 2026–27: Expenditure Profile, Expenditure Budget and Demands for Grants*. New Delhi. Capital expenditure INR 12.2 lakh crore; Ministry of Railways capital outlay INR 2,93,030 crore.
- [25] Central Electricity Authority (2024). *National Electricity Plan, Volume II: Transmission*. Ministry of Power, Government of India, New Delhi.
- [26] AL Circle (2026). *India's aluminium extrusion demand nears 858,000 tonnes*. Industry data series, 2025–26.
- [27] Trade sources on aluminium in Indian rolling stock (2025–26), including *Mobility Outlook* on Titagarh's aluminium metro and Vande Bharat coach programme, and reported tendering for aluminium-bodied Vande Bharat sleeper trainsets.

A Appendix: Technical Framework of ERP Analysis

This appendix provides the technical foundation for the Effective Rate of Protection (ERP) calculations presented in Section 3, utilizing the modified approach established by Pathania & Bhattacharjea (2020) [1]³⁸ to account for domestic price valuations based on Input-Output tables.

A.1 Modified Corden's Approach

We utilize a modified formulation of Corden's measure that adapts to the availability of data from the MOSPI Supply Use Tables (SUTs). We define ERP in sector j as:

$$ERP_j = \frac{VA_{Tj} - VA_{FTj}}{VA_{FTj}}$$

Where VA_{Tj} refers to value added under restricted trade at basic prices, and VA_{FTj} refers to value added under free trade at basic prices.

Using observable SUT data, value added under restricted trade is algebraically represented as:

$$VA_{Tj} = VO_j - \sum_i IC_{ij}$$

Where:

- VO_j : Value of the output of sector j at basic prices.
- IC_{ij} : Intermediate consumption of good i in the production of output j at purchasers' price.

To find the counterfactual value added under free trade (VA_{FTj}), we deflate the ex-post value of output and intermediate inputs by the rate of tariff charged on each product:

$$VA_{FTj} = \frac{VO_j}{(1+t_j)} - \sum_i \frac{IC_{ij}}{(1+t_i)}$$

Where:

- t_j : Nominal ad valorem tariff on the import of final output j .
- t_i : Import-weighted average ad valorem tariff on inputs.

Substituting these into the ERP definition yields the operational formula used for our baseline estimations:

$$ERP_j = \frac{(VO_j - \sum_i IC_{ij}) - \left(\frac{VO_j}{1+t_j} - \sum_i \frac{IC_{ij}}{1+t_i} \right)}{\frac{VO_j}{1+t_j} - \sum_i \frac{IC_{ij}}{1+t_i}}$$

This framework mathematically demonstrates that an Inverted Duty Structure (where $t_i > t_j$, as seen in the Indian downstream aluminium sector under FTAs) directly erodes the domestic value added margin, driving the ERP negative and artificially disprotecting the industry.

³⁸Pathania, K., & Bhattacharjea, A. (2020). *Inverted Duty Structures and the Paradox of Negative Effective Protection in India, 2000-2014*. Foreign Trade Review, 55(2), 139–167.

B Appendix: ASI Measurement — Segmentation Rule, Formulas and Assumptions

This appendix sets out in full the segmentation rule, estimators and assumptions underlying Section 4.1 and the figures that follow it. It is placed here rather than in the body so that the argument can be read without interruption; nothing in Section 4 depends on material that is not stated here.

B.1 The segmentation rule

Following Corden [16], the unit of effective-protection analysis is an **activity**: a process converting traded inputs into a traded output, whose protection depends jointly on the tariff on that output and the tariffs on its inputs. Formally, for an activity j with output tariff t_j and input tariffs t_i over an input set with cost shares a_{ij} (valued at free-trade prices), the effective rate of protection is

$$\text{ERP}_j = \frac{t_j - \sum_i a_{ij} t_i}{1 - \sum_i a_{ij}}. \quad (1)$$

Equation (1) makes two properties explicit that the body of the report relies on. First, $\text{ERP}_j < 0$ whenever $\sum_i a_{ij} t_i > t_j$: a sufficiently protected input drives effective protection on the output negative, which is the inverted-duty-structure case. Second, $\partial \text{ERP}_j / \partial t_i = -a_{ij} / (1 - \sum_i a_{ij})$, so the damage done by an input tariff rises with the input-cost share $\sum_i a_{ij}$ — equivalently, it rises as the share of value added in output falls. This is the formal content of the “transmission mechanism” paragraph in Section 4.7: downstream fabrication carries an input-cost share of 83.1 per cent, so the denominator of (1) is small and any input-tariff wedge is amplified. The operational estimator actually used on Supply–Use data is the modified Corden form given in Appendix A.

The classification rule follows directly. Let $\mathcal{H}$ denote the set of HS headings on which Section 3 estimates effective protection, that is $\mathcal{H} = \{7604, \dots, 7616\}$, and let $h(i)$ be the principal output heading of NIC class i . Then

$$i \in D \iff h(i) \in \mathcal{H}, \quad i \in U \iff h(i) \in \{7601, 7602, 7603\}, \quad (2)$$

giving $U = \{2420\}$ and $D = \{2432, 2511, 2599\}$. Classes whose principal output lies outside HS Chapter 76 — NIC 2710, 2732, 2733 and 3030, whose outputs fall in HS Chapters 85 and 88 — are excluded: they are distinct activities in the sense of (1), facing a different output tariff and a different input vector. The concordance produced by (2) is Table 4.1.

B.2 Notation

For NIC class i and financial year t , the ASI reports

O_{it}	total output	N_{it}	total workers
M_{it}	total inputs	C_{it}	workers employed through contractors
W_{it}	wages and salaries	π_{it}	net profit
L_{it}	total persons engaged	τ_t	effective corporate tax rate in year t

For a segment $S \in \{U, D\}$ all quantities aggregate additively, $X_{St} = \sum_{i \in S} X_{it}$.

B.3 Estimators

Gross value added is the published accounting identity

$$V_{it} = O_{it} - M_{it}, \quad (3)$$

which holds in the source data to within 10^{-6} per cent in every class and year. Value added per person engaged, and the labour-intensity ratio on which Section 4.3 rests, are

$$v_{St} = \frac{V_{St}}{L_{St}}, \quad \lambda_t = \frac{v_{Ut}}{v_{Dt}}. \quad (4)$$

$\lambda_t > 1$ means a rupee of value added created downstream sustains λ_t times the employment of a rupee created upstream. The remaining shares are

$$\omega_{St} = \frac{W_{St}}{V_{St}} \quad (\text{labour share}), \quad c_{St} = \frac{C_{St}}{N_{St}} \quad (\text{contract share}), \quad m_{St} = \frac{M_{St}}{O_{St}} \quad (\text{input-cost share}). \quad (5)$$

Corporate tax and employment yield. Corporate tax is not published by the ASI and is estimated as net profit multiplied by the effective corporate tax rate for the year, with loss-making observations excluded rather than netted off:

$$T_{St} = \tau_t \sum_{i \in S} \pi_{it} \mathbf{1}\{\pi_{it} > 0\}, \quad y_{St} = \frac{L_{St}}{T_{St}}, \quad (6)$$

where y_{St} is persons engaged per unit of corporate tax. The indicator $\mathbf{1}\{\cdot\}$ matters: a negative π would otherwise generate a negative tax, which has no behavioural interpretation. One observation within the retained classes is affected (NIC 2432 in 2013–14).

Index, growth and volatility. With base year $b = 2011-12$,

$$I_{St} = 100 \times \frac{V_{St}}{V_{Sb}}. \quad (7)$$

Because V_{Ut} is volatile, growth is reported on three-year endpoint averages as well as on single endpoints. Writing $A = \{2011-12, 2012-13, 2013-14\}$ and $B = \{2020-21, 2021-22, 2022-23\}$, and $\bar{V}_{SA}$ for the mean of V_{St} over a set A ,

$$g_S = \frac{\bar{V}_{SB}}{\bar{V}_{SA}} - 1, \quad \text{CAGR}_S = \left(\frac{\bar{V}_{SB}}{\bar{V}_{SA}} \right)^{1/9} - 1, \quad (8)$$

the exponent being the nine years separating the midpoints of A and B . Volatility is the sample standard deviation of year-on-year log growth over the twelve clean years,

$$\sigma_S = \text{sd}(\ln V_{S,t+1} - \ln V_{St}). \quad (9)$$

B.4 Construction of each figure

Figure notes in the body carry no derivations. The construction of each is given here. Throughout, $S \in \{U, D\}$, t indexes the financial year, and the base year is $b = 2011-12$. Series are computed on the twelve-year window $t \in [2011-12, 2022-23]$ except where stated.

Figure	Estimator	Construction and domain
4.1	(7)	$I_{St} = 100 V_{St}/V_{Sb}$, both segments, t extended to 2023–24. Current prices; no deflator.
4.2	(4)	$\lambda_t = v_{Ut}/v_{Dt}$. Reference lines: the sample median of $\{\lambda_t\}$, and the exogenous $2.5\times$ benchmark of [4], which is not an estimate of ours.
4.3	(5)	$\omega_{St} = W_{St}/V_{St}$. Numerator and denominator are both monetary, so the series is invariant to the treatment of persons engaged.
4.4	(7)	$I_{it} = 100 V_{it}/V_{ib}$ at class rather than segment level, $i \in \{2420, 2432, 2511, 2599\}$. The horizontal reference is $I = 100$.
4.5	(5)	$c_{St} = C_{St}/N_{St}$. The denominator is total workers N , not persons engaged L : the direct/contract partition is defined only within N .
4.6	(6)	$y_{St} = L_{St}/T_{St}$. Since τ_t is common to both segments it cancels from y_{Dt}/y_{Ut} , so the ratio is invariant to the assumed tax rate.
4.7	—	Four ratios evaluated at $t = 2022-23$, each formed from estimators already defined: $v_U/v_D = 24.97/7.95$; $\omega_D/\omega_U = 38.41/17.59$; $y_D/y_U = 150.83/38.73$; $\sigma_U/\sigma_D = 0.3732/0.1046$. The fourth is oriented upstream-over-downstream because it is upstream value added that is the more variable.

Domain restrictions. All series terminate at 2022–23 under assumption A7 below. Figure 4.1 extends to 2023–24 because the index depends only on V , which is unaffected by the two breaks in that release; every series involving W or the direct/contract partition necessarily stops at 2022–23. Loss-making observations are excluded from T_{St} by the indicator in (6); one observation within the retained classes is affected.

B.5 Two derived results used in Section 4

Amplification of an input-cost wedge. Let $m = M/O$ be bought-in inputs as a share of output value and θ the share of the input bill accounted for by metal. Since $V = (1-m)O$, a proportional rise $\hat{p}$ in the metal price reduces value added by

$$\frac{\Delta V}{V} = -\frac{m\theta}{1-m}\hat{p}, \quad (10)$$

so that $m/(1-m)$ is the factor by which a proportional input-cost change is amplified when it reaches value added. This is equation (1) viewed from the producer’s side: the denominator $1 - \sum_i a_{ij}$ that governs the effective rate is the same value-added share that governs the amplification. With the measured downstream input share $m = 0.8309$ the factor is **4.91**; upstream, where the metal is produced rather than purchased, $\theta \approx 0$ and the effect vanishes.

Metal share of input bill, θ	Value added lost per 1% on metal	Value added lost at a 7.5% wedge
30%	1.48%	11.1%
40%	1.97%	14.7%
50%	2.46%	18.4%
60%	2.95%	22.1%
70%	3.44%	25.8%

θ is not observable in the ASI, which records the input bill in value but not by material. The range spans plausible values for aluminium conversion; firm-level cost data would replace it with a point estimate.

Employment per tonne. Writing Q for tonnes processed, the per-tonne employment ratio decomposes exactly into the per-rupee ratio and the ratio of value added per tonne:

$$\frac{L_D/Q_D}{L_U/Q_U} = \underbrace{\frac{L_D/V_D}{L_U/V_U}}_{\lambda = 3.14} \times \underbrace{\frac{V_D/Q_D}{V_U/Q_U}}_{1.81} = 5.7. \quad (11)$$

Both factors exceed unity, so the two normalisations reinforce rather than offset one another. Tonnage is not an ASI variable: Q_U is installed primary capacity for FY24 and Q_D an industry estimate of 2.0–2.5 Mt, which is why Section 4.3 reports the per-tonne figure as a range of 5.1–6.4 and treats it as supporting rather than primary evidence. Equation (11) also disposes of the much larger ratios sometimes quoted: a figure near 35:1 requires counting the direct workforce of the primary smelters alone against the direct, indirect and unorganised workforce downstream, and the two mismatches together account for the whole of the difference.

B.6 Apportionment, and why it does not affect the results

None of the classes in U or D is exclusive to aluminium. The natural objection is that each should be scaled by an aluminium share $\alpha_i \in (0, 1]$, giving $V_i^{Al} = \alpha_i V_i$ and $L_i^{Al} = \alpha_i L_i$. Applying a coefficient derived from output shares to an employment variable embeds the *industry technology assumption* — that aluminium activity within a class has the same factor intensity as the rest of that class — as against the *commodity technology assumption*, under which each product carries its own factor requirements wherever it is made [23]. The choice is consequential in general. Here it is not, for a reason that can be stated exactly rather than simulated.

Proposition. *For any apportionment vector $\alpha = (\alpha_i)_{i \in D}$ with $\alpha_i > 0$, the apportioned segment productivity $v_D(\alpha)$ is a weighted average of the class productivities $v_i = V_i/L_i$, and is therefore bounded by their range irrespective of α .*

Proof. Writing $w_i(\alpha) = \alpha_i L_i / \sum_{j \in D} \alpha_j L_j$,

$$v_D(\alpha) = \frac{\sum_{i \in D} \alpha_i V_i}{\sum_{i \in D} \alpha_i L_i} = \sum_{i \in D} \frac{\alpha_i L_i}{\sum_{j \in D} \alpha_j L_j} \cdot \frac{V_i}{L_i} = \sum_{i \in D} w_i(\alpha) v_i, \quad (12)$$

with $w_i(\alpha) > 0$ and $\sum_i w_i(\alpha) = 1$. Hence $\min_i v_i \leq v_D(\alpha) \leq \max_i v_i$. Since U is a single class, v_U is exactly invariant to α_U , which cancels in (4). Therefore

$$\frac{v_U}{\max_{i \in D} v_i} \leq \lambda(\alpha) \leq \frac{v_U}{\min_{i \in D} v_i} \quad \text{for every } \alpha. \quad \blacksquare \quad (13)$$

The bound, evaluated. In 2022–23 the class productivities are $v_{2432} = 7.43$, $v_{2511} = 7.38$ and $v_{2599} = 8.30$ lakh per person, against $v_U = 24.97$. Substituting into (13),

$$3.01 \leq \lambda(\alpha) \leq 3.38 \quad \text{for any apportionment whatever,}$$

against the reported undiscounted value of 3.14. The result requires no assumption about the magnitude of any α_i — only that the shares be positive. It also explains why the 48-scenario grid reported in Section 4.9 moves λ so little: every one of those scenarios necessarily lies inside (13).

What is *not* invariant. The same algebra shows that *levels* carry no such protection. The cross-segment employment multiple $L_D(\alpha)/L_U(\alpha) = \sum_i \alpha_i L_i / (\alpha_U L_U)$ is homogeneous of degree one in α and ranges from 0.81 to 3.07 across plausible vectors. This replicates, in a different setting, Bohlin and Widell’s finding that the technology assumption matters greatly for technical coefficients and little for factor-content applications [21]. Accordingly the report quotes intensity ratios, shares and per-rupee statistics throughout, and makes no cross-segment comparison of employment levels.

B.7 Assumptions

The following are stated so that each can be checked or contested independently.

- A1. Partition criterion.** Segments are defined by (2): a class is downstream if and only if its principal output lies in HS 7604–7616. This is Corden’s activity criterion, not a process-stage criterion; the material-flow literature separates semi-fabrication from fabrication, and the consequence of adopting that alternative is reported in Section 4.9.
- A2. Nominal valuation.** All values are at current prices with no deflator applied. Growth rates are therefore nominal and overstate real expansion; comparisons between segments are unaffected, since both are measured on the same basis.
- A3. Principal-product classification.** The ASI assigns an establishment wholly to the NIC class of its principal product. Because NIC 2420’s explanatory note includes semi-manufacturing, foil and wire drawing, output of HS 7604–7608 produced within integrated smelters is recorded upstream. This biases every comparison in the report *against* its own conclusion; no correction is applied.
- A4. No apportionment by metal.** Class aggregates are used undiscounted, so segment totals measure the registered manufacturing space within which aluminium activity sits, not a pure aluminium account. Appendix B.6 shows the reported ratios are bounded irrespective of any apportionment.
- A5. Uniform tax rate.** The effective corporate tax rate τ_t in (6) is applied uniformly across classes within a year. Differences in effective rates between segments are not observable in the published data.
- A6. Loss exclusion.** Loss-making observations are excluded from the tax series rather than netted off, per (6).
- A7. Terminal year.** 2023–24 is excluded from all stated statistics. Reported wages fall in every class in that year while output and persons engaged rise, and the identity $N_{it} = \text{directly employed} + \text{contract fails}$ with a residual of 178,591 workers (19.6 per cent of the total), having held exactly in each of the twelve preceding years.

A8. Frame coverage. The ASI covers registered factories only — units with 10 or more workers using power, or 20 or more without. Downstream aluminium fabrication has a substantial unincorporated segment outside this frame; upstream smelting has none. The downstream aggregate is therefore an undercount to an extent the published data cannot quantify.

A9. Missing classes. NIC 2512 (tanks, reservoirs, gas containers) is not disaggregated in the ASI All-India summary, so HS 7611 and 7613 — two of the thirteen ERP lines — have no ASI counterpart and are absent from the downstream aggregate.

Assumptions A3, A8 and A9 all run in the same direction: each inflates the measured upstream segment or deflates the measured downstream one. The gaps reported in Section 4 are accordingly lower bounds.

B.8 Robustness

Every quantitative claim in Section 4 was tested against the objections a reader is likely to raise. The tests are set out here rather than in the body because several of them are only interesting if the answer is unexpected, and none of them is. Two are findings in their own right and are flagged as such.

Sign consistency: nothing depends on the year chosen. Across the twelve clean years, and without a single exception, downstream fabrication is the more labour-intensive segment ($\lambda_t > 1$); the ratio exceeds the Ministry of Mines' $2.5\times$ lower bound; the downstream wage share exceeds the upstream wage share ($\omega_{Dt} > \omega_{Ut}$); downstream persons engaged exceed upstream ($L_{Dt} > L_{Ut}$); and downstream employment per unit of estimated tax exceeds upstream ($y_{Dt} > y_{Ut}$). Twelve out of twelve on all five. This is the strongest single robustness result in the section: no headline claim is an artefact of the year selected.

Terminal year. Recomputing λ to alternative terminal years gives 3.17, 3.08, 3.75, 5.81, 3.14 and 3.05 for 2018–19 through 2023–24. The value reported in the body, 3.14, sits near the bottom of that range — which is the answer to any suggestion that the terminal year was chosen to flatter.

Base year. Single-endpoint growth comparisons *are* sensitive to the base year, because V_{Ut} is volatile; the sign of the growth differential can be moved by the choice. This is why growth is also reported on the three-year endpoint averages of (8), which are stable: upstream 105.6 per cent against downstream 73.0 per cent, compound rates of 8.34 and 6.28 per cent. Both bases give the same ordering and a similar gap, and no claim in Section 4 rests on the single-endpoint comparison alone.

Leave-one-out. Dropping each downstream class in turn moves the 2022–23 labour-intensity ratio to 3.13 (without 2432), 3.04 (without 2511) and 3.38 (without 2599), against 3.14 for the full basket. No single class drives the result.

The process-taxonomy objection. Section B.1 groups semi-fabrication with fabrication on Corden's criterion, whereas the material-flow literature separates them as distinct process steps [19, 20]. Because NIC 2432 is the semis-and-casting class, the objection can be settled by reassigning it and recomputing. Doing so moves λ from 3.14 to 2.77,

the wage-share gap from 20.8 to 19.6 percentage points, and the employment-yield ratio from 3.89 to 3.31. Every conclusion survives; only the magnitudes move, and modestly. The reason is visible in the underlying figure: NIC 2432's own value added per person engaged is INR 7.43 lakh, close to the INR 7.99 lakh of the remaining fabrication classes and nowhere near the INR 24.97 lakh of primary metal. On the measure that matters here, semi-fabrication behaves like fabrication rather than like smelting, whichever taxonomy one prefers — which is a more durable position than winning the classification argument would have been.

Basket width. Earlier drafts used a wider downstream basket, adding NIC 2710 (motors, generators, transformers) and 2732 (insulated wire and cable). Those classes fail the rule of (2) — their outputs are HS Chapter 85, a different activity protected by different tariffs — but the findings do not depend on excluding them. On the wide basket λ is 2.61 rather than 3.14, the wage-share gap 18.9 rather than 20.8 percentage points, and the employment-yield ratio 2.91 rather than 3.89. Every direction is unchanged; the magnitudes move *against* the wider basket. The narrow basket is preferred not because it is more favourable but because it is the only one consistent with the HS 7604–7616 scope on which the rest of the report is built.

Summary. Table B.1 collects the results. The reported specification is the first row of each block.

Table B.1: Sensitivity of the labour-intensity ratio λ and associated statistics

Specification	λ	Wage-share gap	Jobs per rupee of tax
<i>Reported specification</i>			
Upstream 2420; downstream 2432, 2511, 2599	3.14	20.8 pp	3.89
<i>Terminal year</i>			
2018–19	3.17	—	—
2019–20	3.08	—	—
2020–21	3.75	—	—
2021–22 (metal-price peak)	5.81	—	—
2023–24 (excluded, shown for reference)	3.05	—	—
<i>Leave-one-out on the downstream basket</i>			
Excluding 2432	3.13	—	—
Excluding 2511	3.04	—	—
Excluding 2599	3.38	—	—
<i>Alternative segment definitions</i>			
Material-flow reading: 2432 grouped upstream	2.77	19.6 pp	3.31
Wide basket: adding 2710 and 2732	2.61	18.9 pp	2.91
<i>Apportionment</i>			
Analytic bound over all α , eq. (13)	3.01–3.38	—	—
48-scenario grid	3.19–3.21	—	—

Source: our calculations from ASI All-India summary results, 2022–23 unless otherwise stated. “—” denotes a statistic not recomputed for that variant because it is not at issue. Every row preserves the sign and the order of magnitude of the reported specification; the lowest value anywhere in the table, 2.61, still exceeds the Ministry of Mines’ $2.5\times$ lower bound.

C Appendix: Supplementary Figures

For consolidated reference, all figures presented in the body of the report are reproduced together below, in the order in which they appear. All charts use a greyscale palette with hatch patterning so that every series remains legible in black-and-white printing or photocopying.

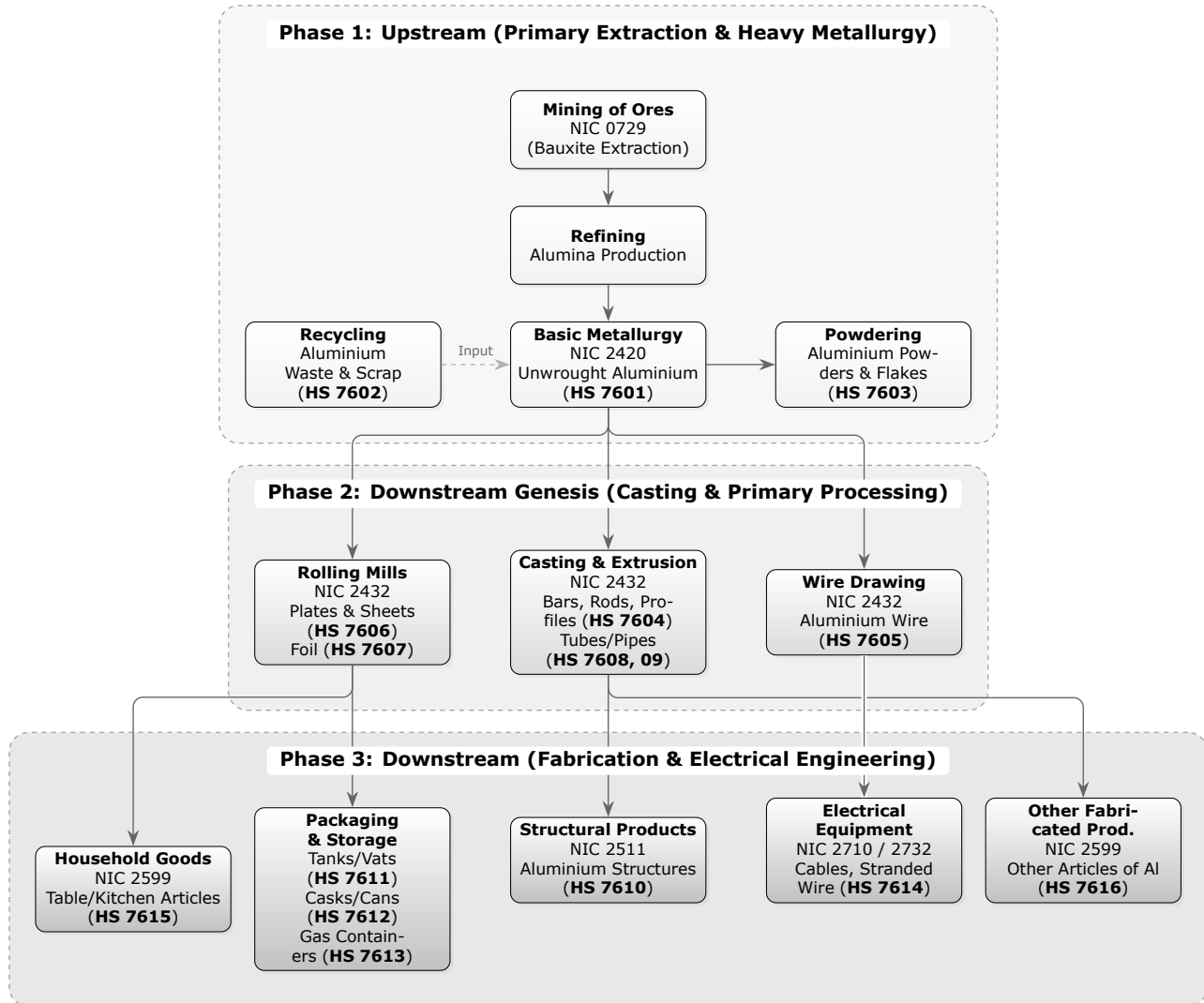
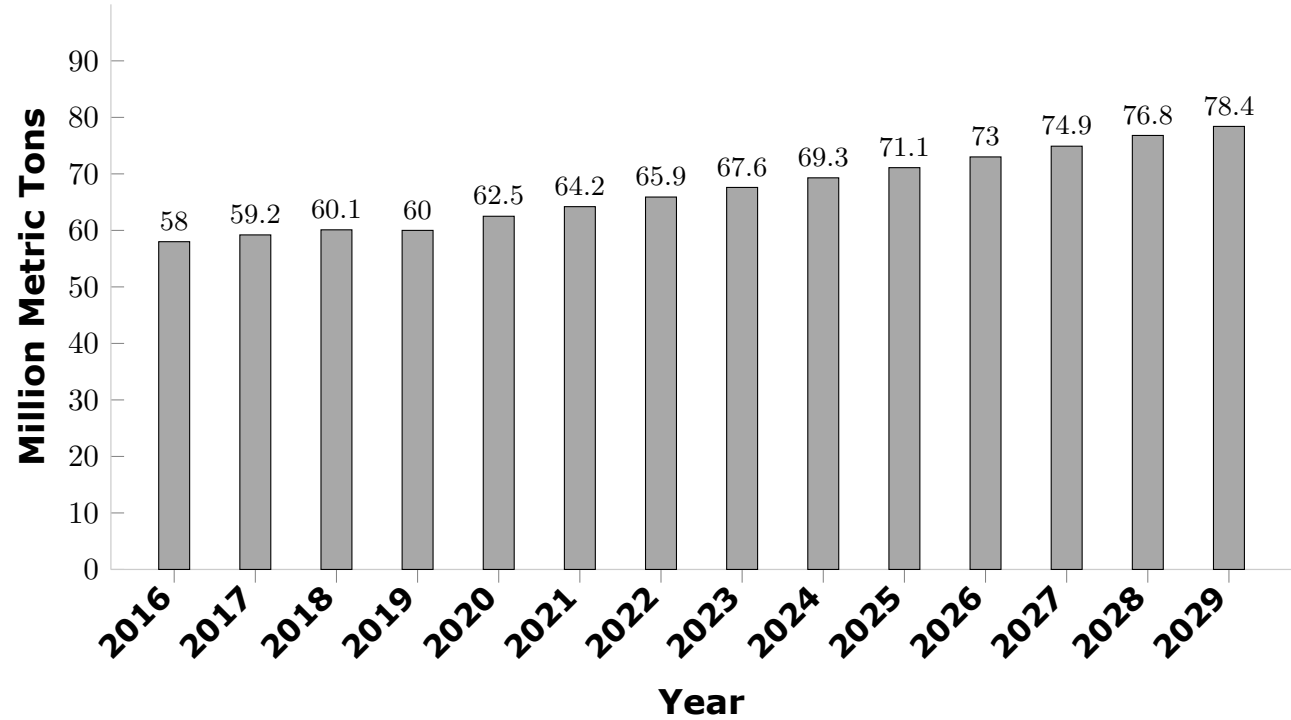


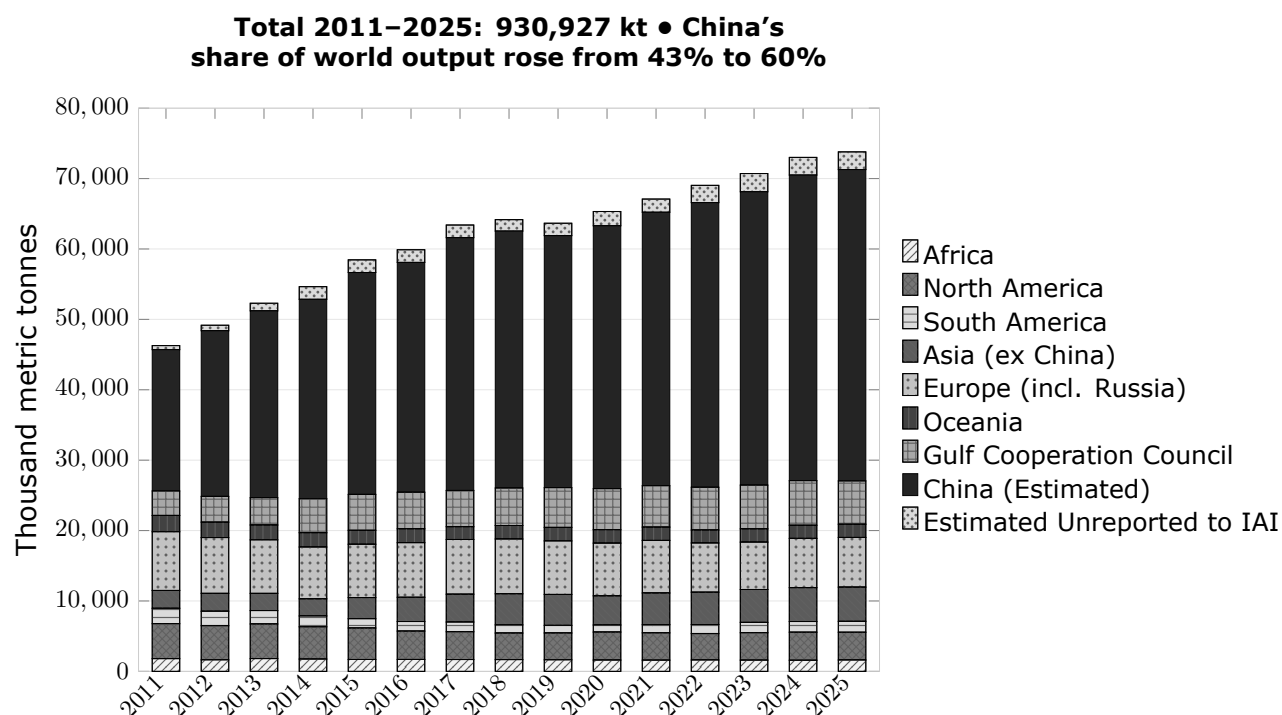
Figure C.1: The Aluminium Value Chain, tracking industrial operations and HS Code progression. (reproduced for reference)

Figure C.2: Global Primary Aluminum Consumption and its Forecast (in million metric tons) (reproduced for reference)



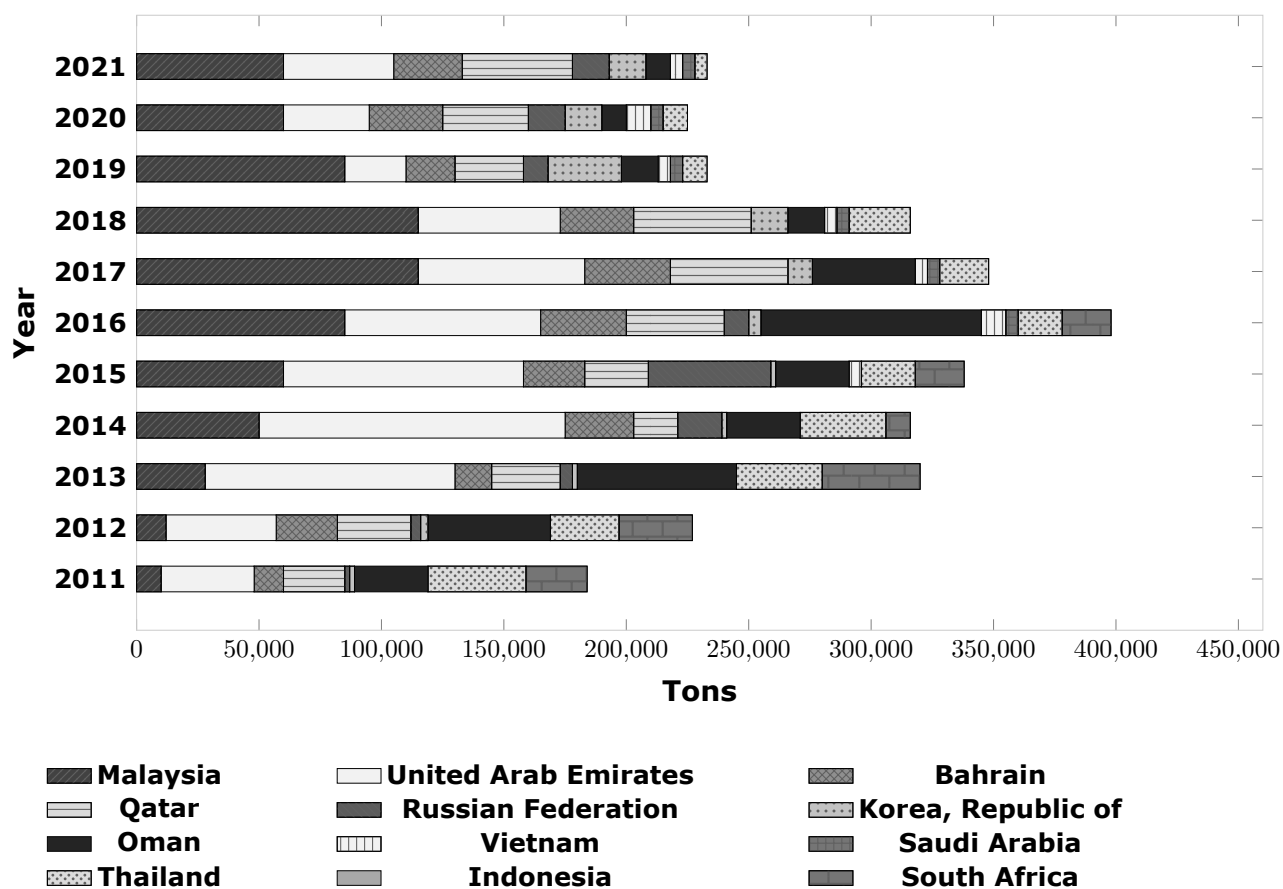
Source: Statista

Figure C.3: Global Primary Aluminium Production by Region, 2011–2025 (thousand metric tonnes) (reproduced for reference)



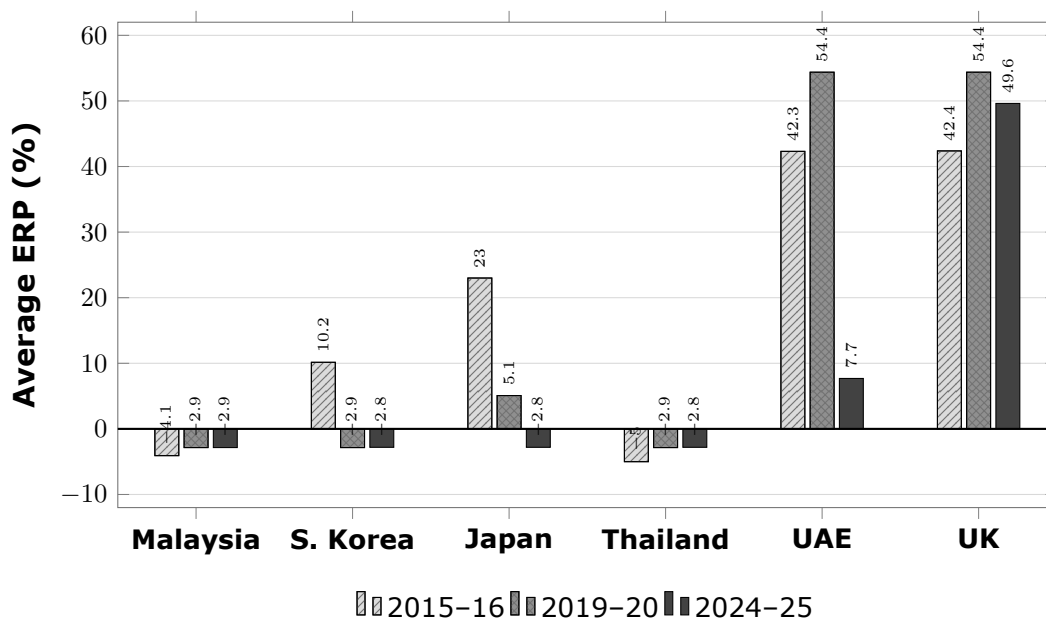
Source: International Aluminium Institute (IAI); annual totals of monthly reported primary aluminium production. Data for Jan–May 2026 are available but omitted for year-on-year comparability. From 2020 onward the IAI reports European output on a combined “Europe (incl. Russia)” basis; earlier “Western & Central” and “Russia & Eastern” figures are aggregated here for a consistent series.

Figure C.4: Top Exporting Countries of Primary Aluminum to India in 2021 (Tons)
(reproduced for reference)



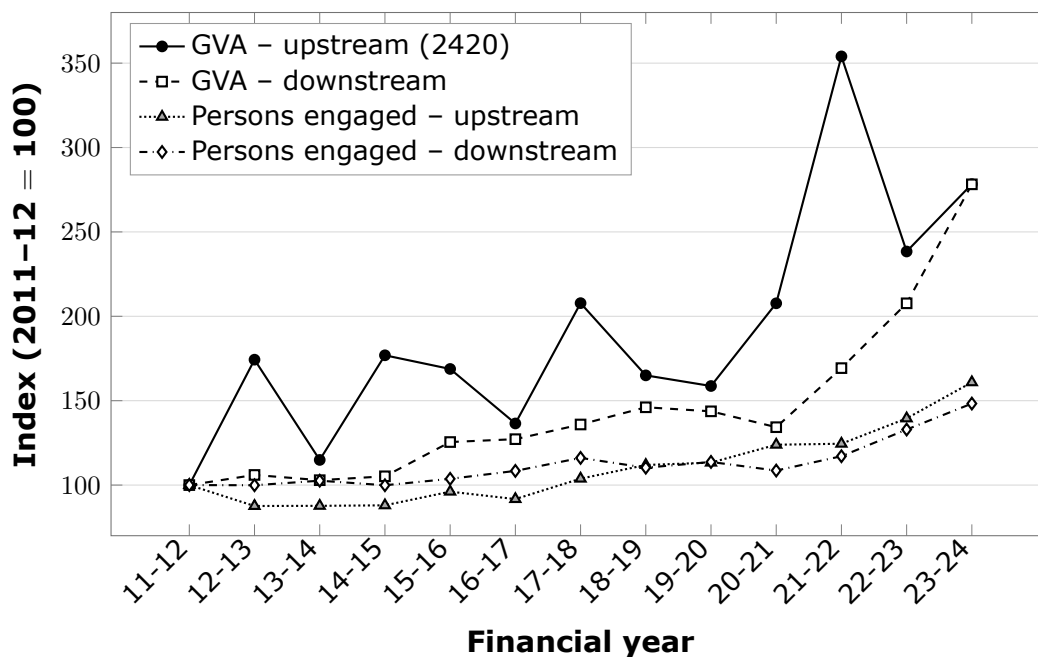
Source: ITC Trade Map

Figure C.5: Average Effective Rate of Protection by partner and period (reproduced for reference)



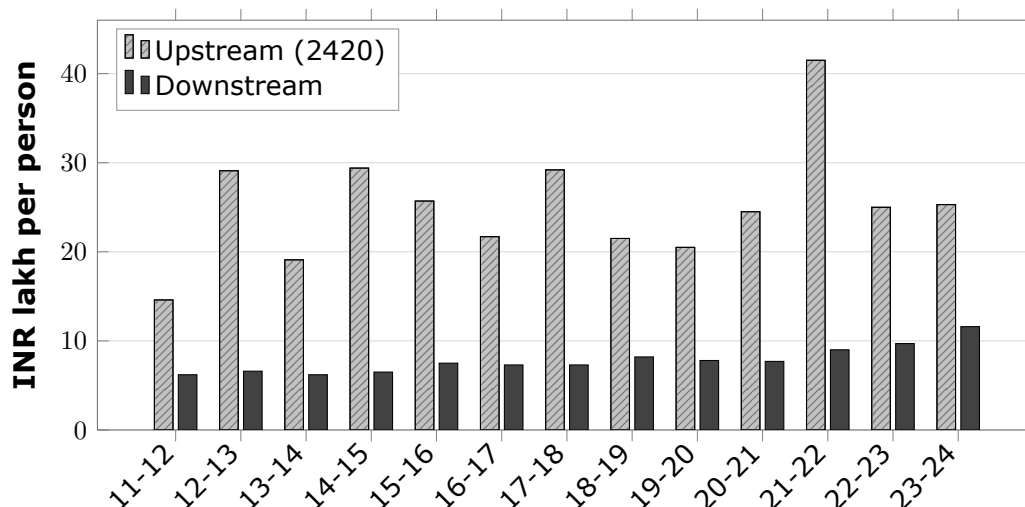
Source: Our estimates from MOSPI Supply Use Tables and partner tariff schedules. Bars below the zero line denote negative effective protection. The UK series is a non-preferential (MFN) benchmark.

Figure C.6: Gross Value Added and Persons Engaged, index 2011-12 = 100 (reproduced for reference)



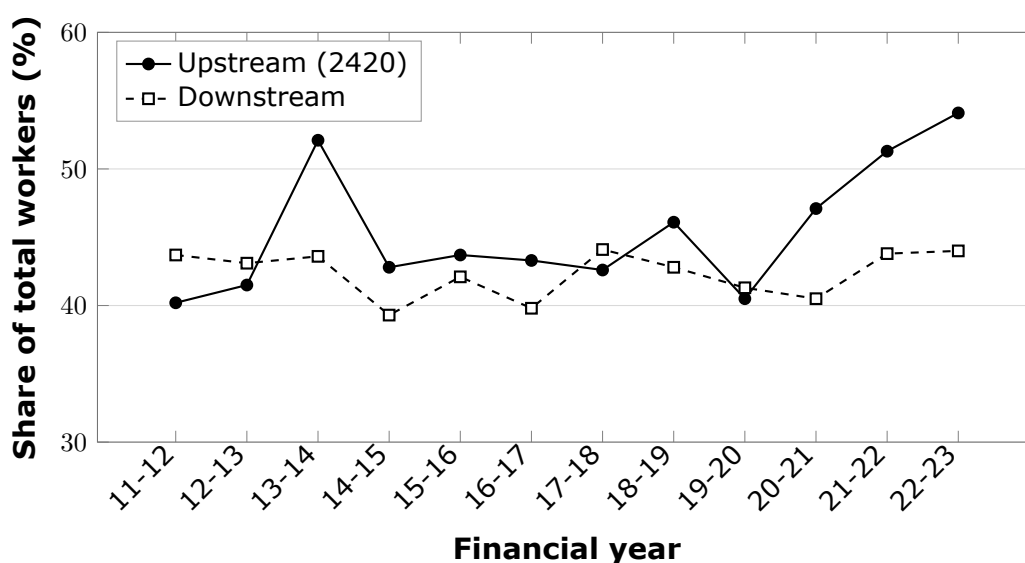
Source: Our calculations from ASI All-India summary results. GVA at current prices (nominal; no deflator applied). Upstream = NIC 2420; downstream = NIC 2432, 2511, 2599, 2710, 2732, 3030.

Figure C.7: Gross Value Added per person engaged (INR lakh, current prices)
(reproduced for reference)



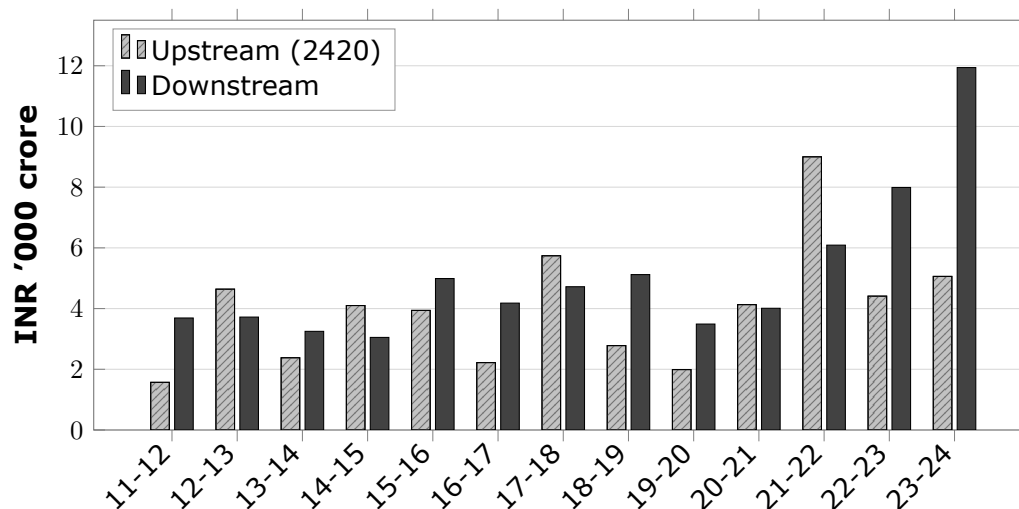
Source: Our calculations from ASI All-India summary results. A *lower* bar denotes greater labour intensity—more persons engaged per unit of value added.

Figure C.8: Workers employed through contractors as a share of total workers (per cent) (reproduced for reference)



Source: Our calculations from ASI All-India summary results. **2023–24 is omitted:** in that year the published directly-employed and contract-worker counts do not sum to total workers (an aggregate residual of 193,626 workers across the eight classes examined), whereas the identity holds exactly in every year from 2011–12 to 2022–23. The series is therefore truncated rather than presented on an inconsistent basis.

Figure C.9: Estimated corporate tax contribution (INR thousand crore) (reproduced for reference)



Source: Our calculations from ASI All-India summary results. Corporate tax estimated as net profit $\times$ the effective corporate tax rate for each year; loss-making observations excluded rather than netted (see Section 4.1).